9 JANUARY 1968
Superseding
MIL-C-9401A(ASG)
31 May 1955

MILITARY SPECIFICATION

CANOPY, PARACHUTE, RING SLOT, GENERAL SPECIFICATION FOR CONSTRUCTION OF

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general construction requirements of a parachute canopy which is characterized by concentric rings of cloth separated from each other by slots and designated as the ring slot parachute canopy.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-295	Thread, Nylon
CCC-T-191	Textile Test Methods

Military

MIL-W-530	Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Colors
MIL-W-4088	Webbings, Textile, Woven Nylon
MIL-C-4232	Cord, Rayon, Without Core, Braided
MIL-T-5038	Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-C-5040	Cord, Nylon
MIL-T-5237	Tape,Textile, Rayon
MIL-T-5608	Tape, Parachute Canopy, Textile, Nylon
MIL-P-5610	Parachute and Parachute Component Parts, Packaging and Packing Procedures for Domestic and Overseas Shipment of
MIL-W-5625	Webbing, Textile, Nylon, Tubular
MIL-T-5661	Tape and Webbing, Textile Cotton Reinforcing, Woven
MIL-W-5665	Webbing, Textile, Cotton Warp

FSC 1670

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MIL-I-6903	Ink, Marking (for Parachutes and Other Textile Items)
MIL-C-7020	Cloth, Nylon, Parachute
MIL-C-7350	Cloth, Nylon, Parachute, Cargo
MIL-C-7515	Cord, Nylon, Coreless
MIL-C-8021	Cloth, Parachute, Nylon, Cargo and Deceleration
MIL-M-46032	Mildew-Resistant Treatment (for Fabrics) Copper Processes

STANDARDS

Federal

FED. STD. No. 751	Stitches, Seams, and Stitchings

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-849	Inspection Requirements, Definitions and Classification of Defects for Parachutes

(Copies of specifications, standards, drawings, and publications required by suppliers in connections with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. - This specification makes provisions for preproduction inspection (see 4.3).

3.2 Component parts. - The component parts of the canopy (see figure 1) shall consist of the drag surface and the suspension lines. For the purpose of this specification, the drag surface shall be defined as the fabric body of the parachute which presents the resistance surface to the air.

3.3 Selection of specifications and standards. - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Detail specification. - The detail specification referenced herein shall be as specified by the procuring activity and may take the form of a specification, drawing, or exhibit or a combination thereof. The requirements specified in the detail specification shall be applicable as requirements of this specification. Where the detail specification and this specification conflict, the detail specification shall take precedence (see 6.2).

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3.5 Materials. - The materials used in the construction of the canopy shall be as specified in the detail specification and shall not necessarily be limited to the material specifications referenced herein. Unless otherwise specified in the material specifications the cotton webbing used in the construction of the canopy shall be mildew-proofed in accordance with type III, Method 1 of MIL-M-46032. Staining of material during the mildew-proofing is acceptable in lieu of dyeing if olive-drab color is required.

3.5.1 Thread. - The thread used in the construction of the canopy shall conform to type I or II, class 1 of V-T-295. The size of the thread shall be as specified in the detail specification.

3.6 Design and construction. - The design and the construction of the canopy shall conform to the applicable requirements specified herein. The dimensions and the over-all proportion of the canopy (including specific values for the applicable dimensions indicated by letters on figures 2 through 29) and the specific style of construction shall be as specified in the detail specification. Measuring operations shall be performed as specified in 3.11.

3.6.1 Gores. - The number of gores in the drag surface shall be the exact number specified in the detail specification. Each gore shall consist of a cloth and webbing grid (trapezoidal sections in the triangular shapes shown on figure 2 or 3) assembled as specified in the detail specification.

3.6.2 Sections. - Whenever possible, the dimensions, including the necessary allowance for seams, in section A on figure 2 or 3 shall be selected to utilize full, half, third, or quarter sizes of standard width material. The number of sections and their dimensions without seam allowances shall be as specified in the detail specification. The sections shall have sufficient seam allowance so that all layers of the seams and overlaps will be caught by all rows of stitching in the radial seams of the drag surface and the hem seams of the sections. The material for the sections may be selected from MIL-C-7020, MIL-C-7350, and MIL-C-8021.

3.6.3 Reinforcement bands. If required by the detail specification, the attachment of the reinforcement bands shown on figure 4 shall conform to either figure 5 or 6. The specific location of the bands on the hem seams of the sections shall be as specified in the detail specification. The ends of the reinforcement band shall overlap 4 inches, and the rows of stitching shall be continuous across the overlap. The lap joints shall be evenly distributed across the drag surface. The material for the reinforcement bands may be selected from MIL-W-530, MIL-W-4088, MIL-T-5038, MIL-T-5237, MIL-T-5608, MIL-T-5661, and MIL-W-5665.

3.6.4 Assembly of the drag surface. - When required, the drag surface shall be assembled by joining the sections into complete rings with basting stitches or any other suitable means and then folding and, if required, reinforcing the edges of the rings. Vertical tapes, if required, shall be added and one or two radial tapes shall be attached. When a single radial tape is required, it shall be placed on the outside of the canopy. The style used in joining the sections and the number of rows of stitchings in the radial seams shall be as specified in the detail specification and shall be in accordance with one of the styles shown on figures 7 and 8.

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3.6.5 Radial tape. - The length of the radial tape extension beyond the skirt shall be determined by the style of the suspension line attachment. The material for the radial tapes may be slected from MIL-W-530, MIL-W-4088, MIL-T-5038, MIL-T-5237, MIL-T-5608, MIL-W-5625, MIL-T-5661, and MIL-W-5665.

3.6.6 Radial seam reinforcement. - The radial seam reinforcement, if required, shall be centered in the radial seam between the radial tapes and shall be caught by the maximum number of rows of stitching. The material for the radial seam reinforcements shall be as specified in 3.6.5.

3.6.7 Vertical tape. - When vertical tape is required by the detail specification, the style of attaching the vertical tape to the sections by stitching shall be as specified in the detail specification and shall be in accordance with one of the styles shown on figures 9 and 10. The detail specification will specify whether single or double vertical tapes shall be used. When a single vertical tape is required, the tape shall be on the outside of the canopy when the canopy is inflated. The material for the vertical tapes shall be as specified in 3.6.3.

3.6.8 Skirt. - The construction of the skirt shall be as specified in the detail specification and shall be in accordance with one of the styles shown on figures 11 and 12. When applicable, the free ends of the vertical and the radial tapes shall be folded in or around the skirt hem seam so that no cut edge is exposed after the skirt reinforcement is attached.

3.6.8.1 Skirt reinforcement. - The ends of the skirt reinforcement shall overlap 6 inches. The number of pieces used shall be as specified in the detail specification. When more than one piece is used, the shortest length shall cross a minimum of four gores. The rows of stitching shall be continuous across the overlap. The material for the skirt reinforcements shall be as specified in 3.6.3.

3.6.9 Vent. - The vent shall be in accordance with figures 13 and 14. The vent construction at the radial seam shall be as specified in the detail specification and shall be in accordance with one of the styles shown on figures 15, 16, and 17. When applicable, the free end of the vertical and the radial tapes shall be folded in or around the vent hem seam so that no cut edge is exposed after the vent reinforcement is attached. When required by the detail specification, tucks shall be taken in the vent sections of each gore to provide fullness in the gore. If the construction shown on figure 17 is specified in the detail specification, the binding shall be made of the same material as the sections.

3.6.9.1 Vent reinforcement. - The ends of the vent reinforcement shall overlap a minimum of 4 inches. The rows of stitching shall be continuous across the overlap. The lap joint shall be located so that the ends of the overlap will be secured by the stitching of at least two consecutive vent lines. The material for the vent reinforcement shall be as specified in 3.6.3 and shall be one continuous piece.

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3.6.9.2 Vent line. - The vent lines shall cross the vent in proper rotation and shall not show evidence of twists. Although normally shorter than the constructed diameter of the vent, the free length of the vent line (dimension L_v on figure 13) shall be as specified in the detail specification. The attachment of the vent lines to the radial tapes (see figure 18 or 19) shall be as specified in the detail specification. The material for the vent lines may be selected from MIL-W-4088, MIL-C-4232, MIL-C-5040, MIL-W-5625, and MIL-C-7515. Unless otherwise specified in the detail specification, the vent lines shall be of the same material as the suspension lines.

3.6.10 Bridle centering lines. - When required by the detail specification, the bridle centering line shall be attached to the topmost vent line as shown on figure 20. The dimensions L_x and L_y on figure 20 shall be as specified in the detail specification. Unless otherwise specified in the detail specification, the material for the bridle centering line shall be the same as the vent lines.

3.6.11 Pocket band. - When required by the detail specification, pocket bands shall be attached to the canopy as shown on figure 21. The style of attachment, the number of pocket bands, and the dimensions L_a and L_b on figure 21 shall be as specified in the detail specification. The material for the pocket bands may be selected from MIL-W-530, MIL-W-4088, MIL-C-4232, MIL-T-5038, MIL-C-5040, MIL-T-5237, MIL-T-5608, MIL-W-5625, MIL-T-5661, MIL-W-5665, and MIL-C-7515.

3.6.12 Suspension line. - The free length of the suspension line (see dimensions L_s on figures 22 through 27) shall be as specified in the detail specification. Sufficient length on both ends of the suspension line shall be allowed for the style of suspension line loop required. If style A is used, the end of the suspension line shall be taper cut and inserted inside the lining as shown on figure 28. Suspension lines shall be continuous and without splices or knots. Suspension lines damaged during the process of canopy manufacture shall be replaced with lines fabricated by the same line manufacture. The material for the suspension lines may be selected from MIL-W-4088, MIL-C-4232, MIL-C-5040, MIL-W-5625, and MIL-C-7515.

3.6.12.1 Attachment to drag surface. - The attachment of the suspension line to the drag surface (see figures 22 through 27) shall be as specified in the detail specification. When the attachment loops shown on figures 22 through 25 are used, the knot of the suspension line loop shown on figures 28 through 30 shall be drawn snugly against the loop before sewing. The material for the attachment loops and the radial tape loop reinforcements shall be as specified in 3.6.5.

3.6.12.2 Attachment to load. - The method of attaching the suspension line to the load shall be as specified in the detail specification.

3.7 Stitching. - Stitching shall conform to FED. STD. No. 751. Unless otherwise specified, the stitch type shall be 301 of FED. STD. No. 751. The number of stitches per inch applicable to stitch types shall be as specified in table I.

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TABLE I. Thread size and stitches per inch

Thread size	Straight stitching		Zigzag stitch 2/	
	Type 301	Type 401	Type 304 single throw	Type 308 double throw
E	1/ 7 to 11	1/ 7 to 11	12 to 16	8 to 12
F	1/ 7 to 11		12 to 14	7 to 10
FF	1/ 6 to 9		12 to 14	7 to 10
3-cord	5 to 8			4 to 6
5-cord	4 to 6			3 to 5
6-cord	4 to 6			3 to 5

1/ Stitches not exceeding 16 stitches per inch will be acceptable for a length not exceeding 2 inches in any one place with a minimum distance between locations of 12 inches. More than 16 stitches per inch will be cause for rejection. Where abrupt changes in material thickness occur, a variation of ±2 stitches per inch beyond the specified number for a distance not exceeding 3/4 inch will be acceptable.

2/ When the material thickness is more than doubled, the width of the zigzag stitching may vary ±50 percent or the number of stitches may vary ±2 stitches per inch beyond the number specified for a distance not exceeding 3/4 inch.

3.7.1 Tension. - The tension of the bobbin and the needle threads shall be equal so that the thread locks will lie as close as possible in the center of the layers below the surface. Needles of multiple-needle machines shall be adjusted to provide the same tension from each needle insofar as practicable.

3.7.2 Straight stitching. - Except for stitching made with stitch type 401, straight stitching shall be backstitched not less than 1/2 inch, unless turned under in a hem or held down by other stitching. Unless otherwise specified in the detail specification, the outside row of stitching made with a multiple-needle machine shall be 1/8 ±1/16 inch from the edge of the seam or the material. Multiple rows of straight stitching, when specified in the detail specification, shall be distributed equally across the seam when a single-needle machine is used. Stitching made with stitch type 401 shall be properly and carefully locked where the stitching ends.

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3.7.3 Cross and box stitching. - Cross and box stitching shall be as straight, neat, and as even as possible. Outlines may be used as stitching guides for cross and box stitching. Outlines or any other markings used shall be approved materials to preclude the possibility of damage to the canopy.

3.7.4 Zigzag stitching. - Zigzag stitching (see 6.3.1) that is parallel to straight stitching shall normally not overlap the straight stitching. When only one row of zigzag stitching is specified, it shall be centered. Unless otherwise specified, the tolerance for the length of the rows of zigzag stitching shall be +1/2 or -0 inch. When double-throw zigzag stitching on a single cord is specified, one-quarter length of any stitched row may be pierced by only two needles provided that the cord conforms to MIL-C-5040. The distance between the nearest edge of the zigzag stitching and the edge of the material or seam and the distance between the adjacent edges of the rows of zigzag stitching on the drag surface shall be not less than 1/8 inch. The zigzag stitching on the suspension line (see figure 28 or 29) shall be as specified in the detail specification. The width of the zigzag stitching on the suspension line loop specified on figure 29 shall be 3/16 ±1/64 inch and centered across the two widths of the suspension line so that both outside needle positions pass through the suspension line.

3.7.5 Basting stitches. - Except for size E or lighter nylon thread, basting thread shall be removed.

3.7.6 Automatic sewing machines. - Automatic sewing machines may be used for sewing stitch patterns provided that the size of the pattern, the thread size, and the number of stitches per inch specified in the detail specification are maintained. A minimum of three tie-in, overlapping or back stitches shall be used to secure the ends.

3.8 Seams. - Seams shall conform to FED. STD. No. 751. Seams shall be flat and smooth to preclude pleating, ragged edges, cloth edges not properly caught, and puffiness of fabric between the rows of stitching. Every effort shall be made to produce a fully folded flat seam. Excessive puffiness in the seams shall not be acceptable, because it tends to reduce the fullness in the gores causing stresses in the seams during canopy inflation which may cause damage or failure. Finished fell and hem seams showing raw edges shall be repaired. Seam margins that are less than 1/16 inch, but not beyond the edge of seam or material, will be acceptable for a distance of 2 inches spaced not less than 36 inches apart.

3.9 Splices. - Splices of webbing tapes, and ribbons used in assembling the canopy, except the vent reinforcement web, shall be held to a minimum and shall not exceed the number specified in table II. Only one piece of webbing shall be used in the vent hem. When two or more layers of the same material are used in a seam, one layer shall be continuous across the splice of the other layer. Unless otherwise specified, the ends of the splice shall be made by overlapping the ends for a distance of not less than 4 inches.

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TABLE II. Frequency of splices and repairs

Length of seam (inches)	Number of splices permitted	Number of repairs permitted
0 to 2	0	0
2 to 12	0	1
12 to 72	0	2
72 to 240	1	3
240 to 360	2	4
360 to 480	3	5
480 to 600	4	6
Over 600	1 per 120 inches	1 per 100 inches

3.10 Repairs. -

3.10.1 Broken or skipped stitches. - Broken or skipped stitches in stitch type 301 shall be repaired by adjacent stitching with the stitches extending a minimum of 3 inches (±1 inch) beyond the ends of the stitches being repaired. Broken or skipped stitches in stitch type 304 or 308 shall be repaired by adjacent stitching with the stitches extending a minimum of three stitches beyond the ends of the stitches being repaired. Broken or skipped stitches in stitch type 401 shall be repaired with stitch type 301 and as specified for broken or skipped stitches in stitch type 301. The frequency of the stitching repairs shall not exceed the number specified in table II.

3.10.2 Run-offs. - A run-off shall be repaired by stitching within the seam margin tolerance with the stitches extending a minimum of 1 inch or three stitches, as applicable to the type of stitch, from the start and the end of the run-off.

3.10.3 Less than specified number of stitches per inch. - Where the number of stitches per inch is less than the specified number, repairs shall be made by adjacent stitching, with the repair extending either 1 inch or three stitches, as applicable to the type of stitch, beyond the end of the repair.

3.10.4 Loose stitches. - Loose stitching shall not be removed but shall be reinforced by adjacent stitching. Loose stitches shall be interpreted as those that do not lie smoothly on the fabric when the seam is put under tension.

3.10.5 Tight stitches. - Tight stitches shall be removed and the section shall be restitched. Tight stitches shall be interpreted as those that cause the fabric under the stitches to gather or pucker when the seam is put under tension.

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3.10.6 Darning tears and cuts. - Darning shall be held to a minimum. Holes (see 6.3.3) not exceeding 1/4 inch in diameter and linear tears or cuts not exceeding 1/2 inch may be darned. The darned area of the holes shall extend 3/8 inch beyond the hole on all sides and shall be either a round or a rectangular pattern. The darned area of linear tears or cuts shall form a rectangular pattern extending 1/4 inch beyond each side and each end of the cut or tear. All darns shall be neat and flat and shall not pucker the canopy cloth. The darning thread shall be size A thread or yarns from the canopy and shall match the canopy cloth.

3.11 Measuring operations.

3.11.1 Cloth. - All cloth shall be measured and cut while under substantially no tension. Tightly rolled material shall be relaxed by unrolling and, if necessary, relaxing it for 24 hours to permit the material to return to its original state. Section pattern shall be carefully placed on the cloth so that the small inequities of various cloth widths (within the tolerance specified) are evenly distributed on each side of the pattern. When multiple layers of cloth are cut at the same time, extreme care shall be taken to prevent the cloth from slipping during any part of the cutting operation.

3.11.2 Suspension and vent lines. - The suspension and the vent lines shall be measured under the tensions specified in table III and marked to show the points of attachment to the drag surface and the load. Care shall be exercised in preparing proper line-measuring equipment. Equipment such as pulleys, guides, and reels, if used, shall not induce false loads on the material nor uneven load distribution on multiple line-measuring systems. Reasonable care shall be exercised to maintain uniformity in time control of load applications and control of large changes in temperature and humidity. Lines measured after attachment may have a variation in length of 1 percent in any one canopy when measured under tension in accordance with table III.

TABLE III. Tensile strength and tension

Tensile strength (pounds)	Tension (pounds)
100 to 500	10 ±1/2
500 to 1,000	20 ±1
1,000 and up	40 ±2

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3.11.3 Webbings. - Webbings, tapes, and ribbons used in assembling the drag surface shall be measured with no tension applied during the measuring operations. Very heavy webbings shall not be measured while in the stretch state. (The stretched state of webbing is caused by winding the webbing onto the spool while the webbing is under tension.) A uniform period of time shall be maintained between load applications and marking.

3.11.4 Drag surface. - When finished dimensions are specified, proper allowance shall be provided for stitching take-up to obtain the finished dimensions within the specified tolerances under any condition of humidity. Unless otherwise specified in the detail specification, all finished dimensions shall be measured on the finished drag surface under nominal tension within 15 seconds after application of the nominal tension (see 6.3.4).

3.11.5 Stitches. - The number and the spacing of stitches shall be measured under a tension of not less than 1/2 pound nor more than 1 pound.

3.12 Physical strength. - Unless otherwise specified, the stitching connections (including the suspension line-skirt area and the vent line-vent area) shall withstand an ultimate load of at least 90 percent of the rated strength of the weakest section.

3.13 Tolerances. - Unless otherwise specified, the tolerances for dimensions shall be as specified in table IV.

TABLE IV. Tolerances for dimensions

Dimensions (inches)	Tolerances
1/16	±1/32
Under 2 but over 1/16	±1/16
2 to 10	±1/8
10 to 30	±1/4
30 to 60	±3/8
60 and up	1/

1/ The tolerance is 1 percent.

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3.14 Color. - Unless otherwise specified in the detail specification, the color of the canopy material, excluding the thread, shall be natural color. Unless otherwise specified in the detail specification, the thread used for sewing the canopy shall be in contrasting colors, that is, olive-drab thread on white and extremely light tinted materials and white thread on color materials. The color of webbings, tapes, and cords fully concealed in seams is optional.

3.15 Finish. -

3.15.1 Ends. - The ends of all materials which, in the final assembly, are not covered by other cloth, tapes, webbings, or ribbons shall be treated to prevent fraying. The exposed ends of nylon cords, webbings, tapes, and ribbons shall be cut and seared smoothly and evenly or dipped in a melted mixture of 50 percent beeswax and 50 percent paraffin at a temperature of 180° ±20° F. The ends of suspension lines shall be neatly trimmed back to the stitching, leaving an end which is not longer than 1/4 inch.

3.15.2 Searing. - Searing shall not be permitted after stitching. Any searing permitted shall not leave sharp edges or points that may cause snagging of the canopy cloth. In general, searing shall not be used where the end of the material is sandwiched between two other pieces of material or where the stitching runs continuously over the exposed ends. All finishing shall be done neatly and shall not have any sharp edges or excess melted materials which could cause localized stresses. Whenever searing is used, the finished canopy shall be carefully examined for holes and snags caused by poorly seared materials. One broken warp thread will be allowed in a tape or webbing, provided that the strength of the material is not below specification strength.

3.16 Marking. - The fluid used to mark the canopy shall conform to MIL-I-6903. Unless otherwise specified in the detail specification, contrasting colors shall be used for ease in identification.

3.16.1 Identification of product. - Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The nomenclature of the canopy shall be as specified in the detail specification.

3.16.2 Gore numbers. - Each gore shall be numbered by stamping or stenciling the number of the gore in the right-hand corner on the outside of the lower sections on or adjacent to the skirt. The numbering sequence shall be clockwise when viewing the canopy from the top. The height of the numbers shall be not less than 1/2 inch.

3.16.3 Additional marking. - Each parachute canopy designated for aircraft deceleration shall have the words THIS SIDE OUT stamped or stenciled near the skirt on the outside of the skirt section of each gore. Each extraction parachute canopy procured for Army use shall be stamped or stenciled THIS SIDE OUT on the top and bottom gores only near the skirt, on the outside of the skirt sections. The size of the characters shall be a minimum of 1 inch high.

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3.17 Workmanship. - The canopy, including all materials and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes and defects, accuracy of dimensions and markings, neatness and thoroughness of stitching, and proper assembly of component parts.

3.18 Maintainability. - The canopy design and material should be such that no maintenance is required under normal handling and usage. The canopy design and materials should be such that minor repairs may be accomplished at an advance base or aboard an aircraft carrier.

3.19 Reliability. - Where applicable, contracting documents shall contain specific reliability goals and minimum acceptable standards. The Government reserves the right to maintain surveillance over the contractor's reliability program. The contractor's system of failure prediction, failure mode analysis, design review, test and demonstration, analysis of failure feed-back data, and corrective action are elements that shall be monitored by Government inspection or engineering personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the preformance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - The examination and testing of parachute canopies shall be classified as follows:

(a) Preproduction inspection (4.3)
(b) Quality conformance inspection (4.4)

4.3 Preproduction inspection. - The preproduction inspection samples, consisting of three samples of each stitching connection specified in 3.12 and in the contract (see 6.2), shall be subjected to all of the applicable tests specified herein and in the detail specification. The preproduction inspection shall be performed by the activity and at the test facility specified by the procuring activity (see 6.2).

4.4 Quality conformance inspection. - The quality conformance inspection shall consist of the examination of product specified in 4.5.1.

4.4.1 Rejection and retest. - Canopies which have been rejected may be reworked or replaced to correct the defects, and resubmitted for acceptance. When resubmitting, full particulars concerning previous rejections and the action taken to correct the defect(s) found in the original article shall be

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furnished to both the contractor and the procuring activity, along with the resubmitted article. Canopies rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.4.2 Material strength. - Three 1-yard long samples of each type cloth, cord, ribbon, web, or tape of the lot(s) of material used in the manufacture of the canopy shall be furnished. One spool (or similar unit of issue) of each type of thread or cord used in canopy construction shall also be furnished, and shall be of the same lot used in construction. These samples shall be tested in accordance with CCC-T-191 for the physical properties as given in the applicable specification for the material.

4.5 Inspection methods. -

4.5.1 Examination of product. - Each canopy shall be examined to determine compliance with the requirements specified herein, and in the detail specification, with respect to materials, workmanship, and marking. Inspection requirements and classification of defects shall be in accordance with MIL-STD-849.

4.5.2 Physical strength. - Three samples representative of each stitching connection specified in 3.12 shall be pull tested on a machine of approved type to determine the breaking strength of the connection. The machine used shall be adjusted to operate the pulling jaws, under no load, at a separation rate of 5 inches per minute. Failure of any sample to meet the requirements specified in 3.12 shall be cause for rejection of all the canopies.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. - Unless otherwise specified by the procuring activity (see 6.2), the canopy shall be preserved, packaged, and packed in accordance with type II of MIL-P-5610.

5.2 Marking of shipments. - Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature of the canopy shall be as specified in the detail specification.

6. NOTES

6.1 Intended use. - The ring slot parachute canopy covered by this specification is intended for application where high strength, low opening shock, and stability are of paramount importance, such as the deceleration of aircraft during approach and landing operation, the deceleration of missiles and target aircraft, and the aerial delivery of supplies and equipment.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Detail specification number and responsibility to report in writing, immediately upon discovery, any discrepancies between this specification and the detail specification (see 3.4).

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(c) Specify samples required, location of activity to perform preproduction tests, and location of test facility (see 4.3).
(d) Preservation, packaging, and packing required (see 5.1).

6.3 Definitions. -

6.3.1 Zigzag stitching. - A single-throw zigzag stitch is defined as having two needle drops. A double-throw zigzag stitch is defined as having three needle drops.

6.3.2 Repair. - A repair is the replacement of missing, broken, or removed thread of yarn.

6.3.3 Hole. - A hole is defined as two or more broken yarns in the material in either the warp or the fill, or both.

6.3.4 Nominal tension. - Nominal tension is a pull of 1 percent of the specified strength of the material.

6.4 Marginal indicia. - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - AS
Army - GL
Air Force - 11

Reviewer activities:
Navy - AS
Army - GL
Air Force - 11, 82

User activity:
Navy - MC

Preparing activity:
Air Force - 11

Project No. 1670-0248

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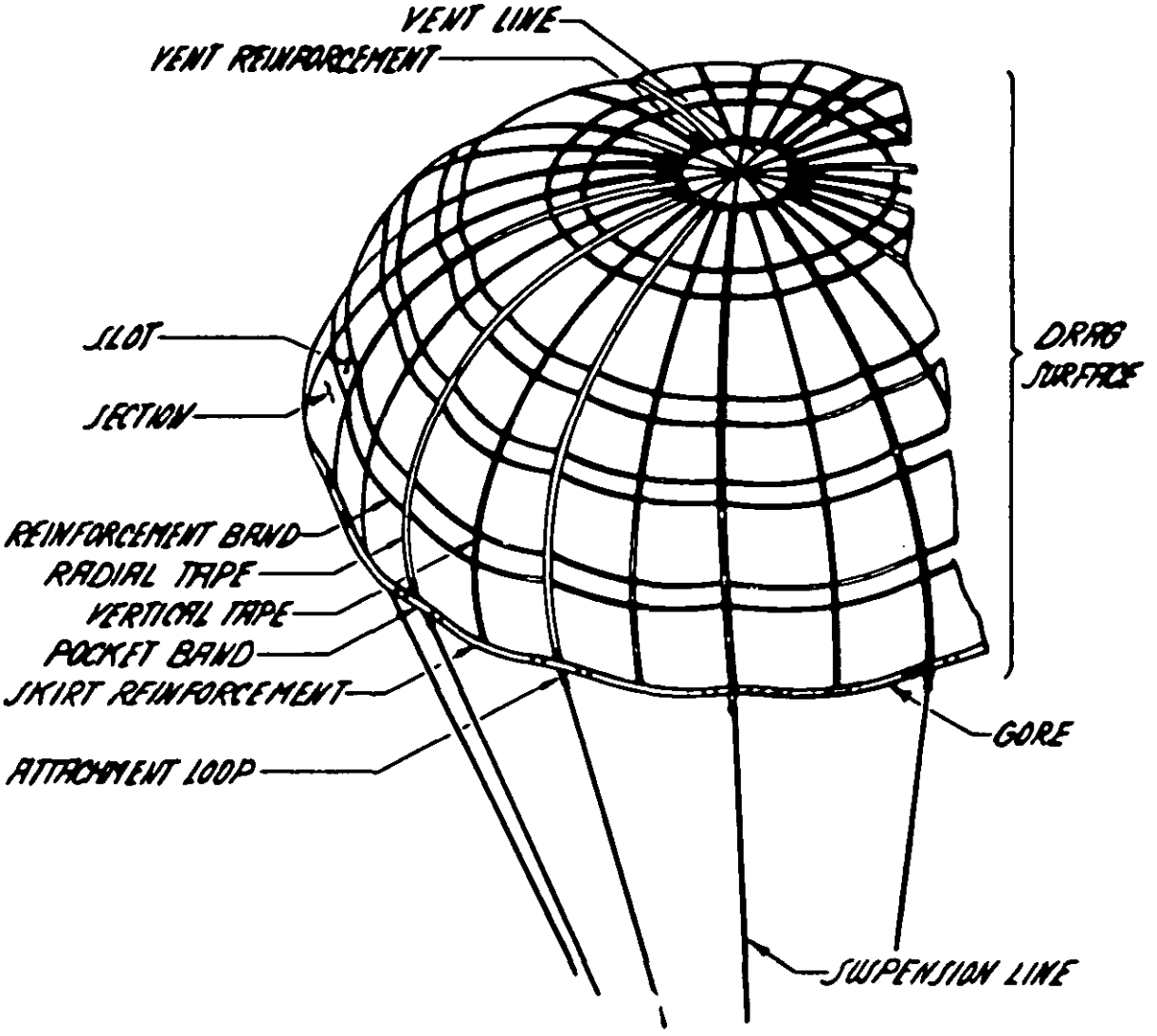


FIGURE 1. Component parts of a ring slot canopy

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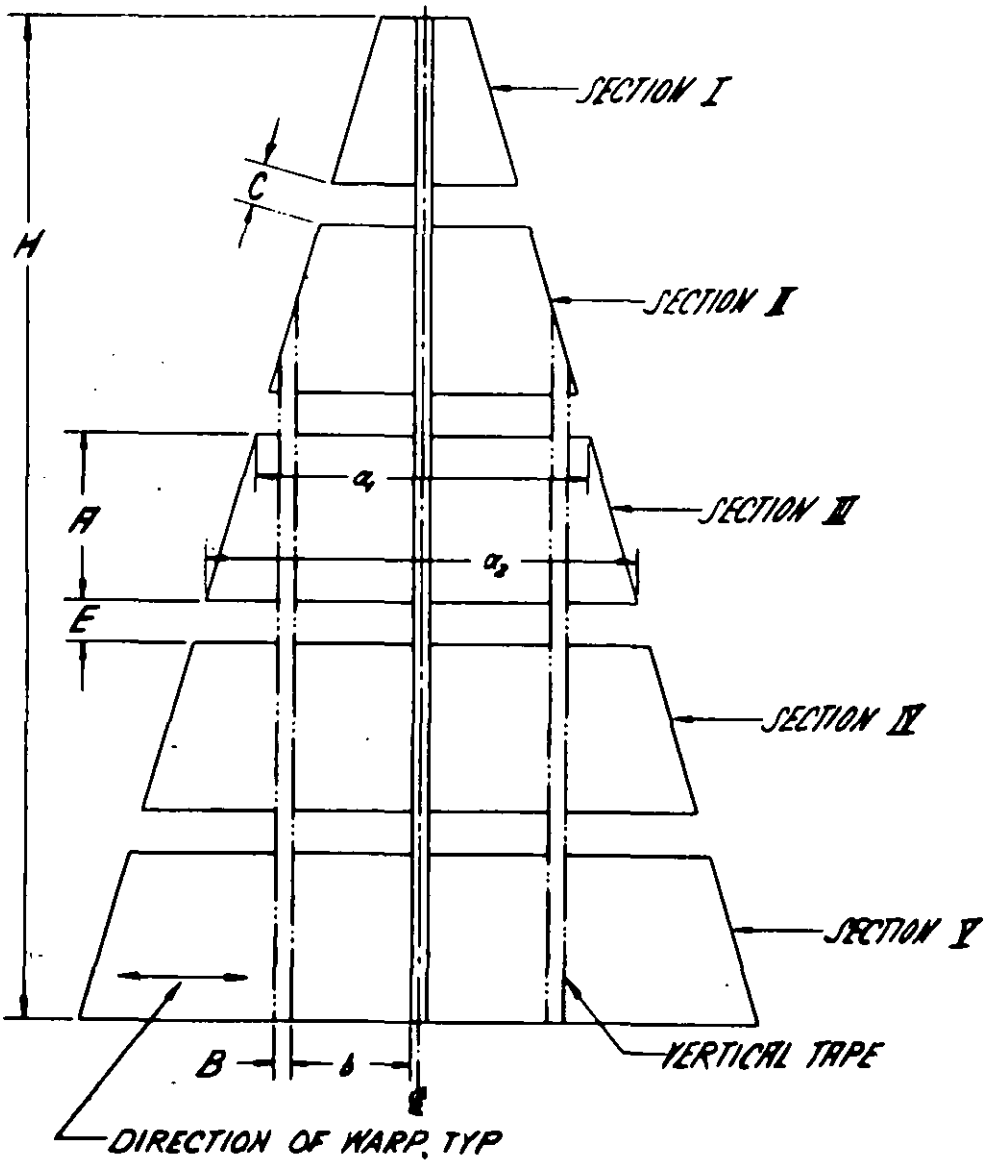


FIGURE 2. Gore configuration, style A

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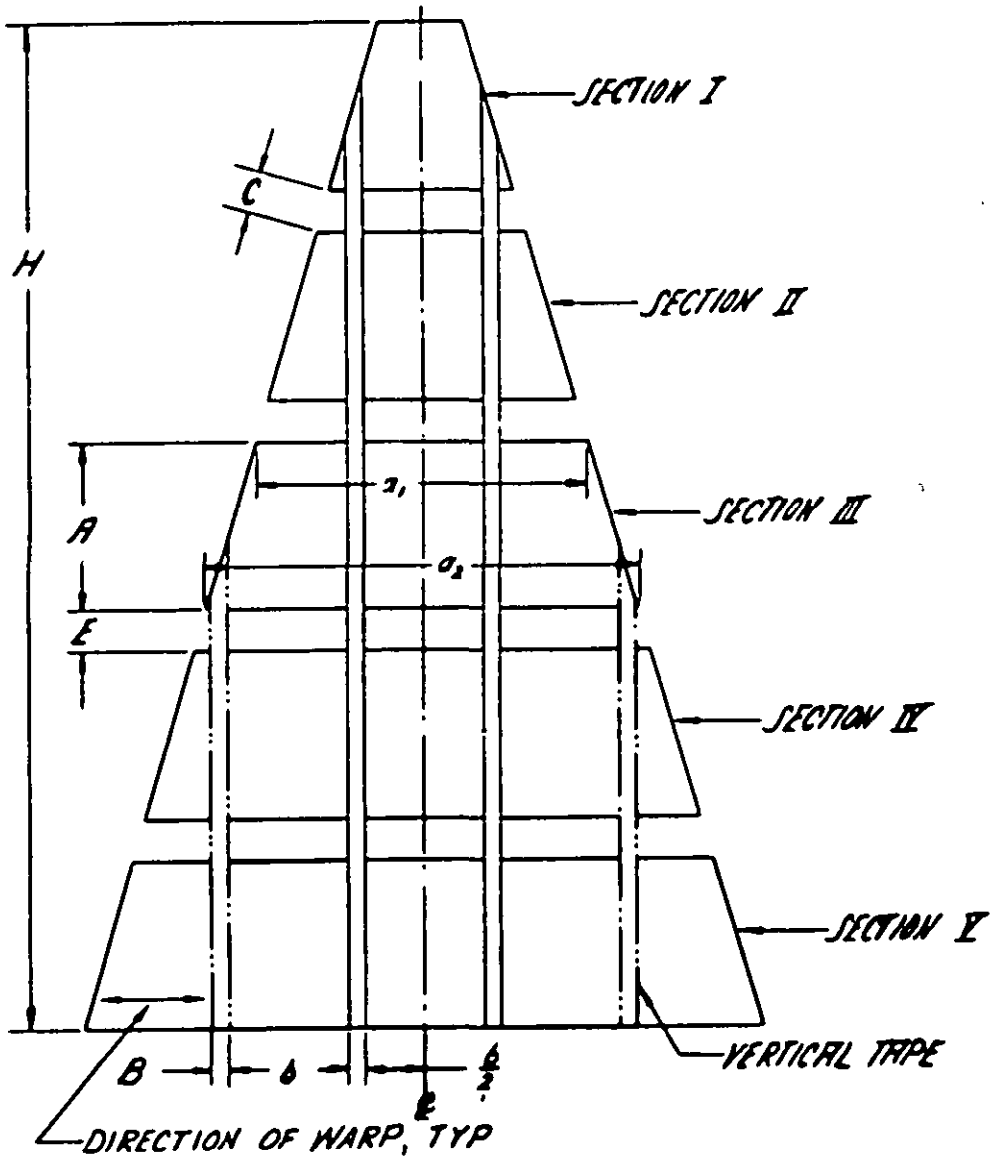


FIGURE 3. Gore configuration, style B

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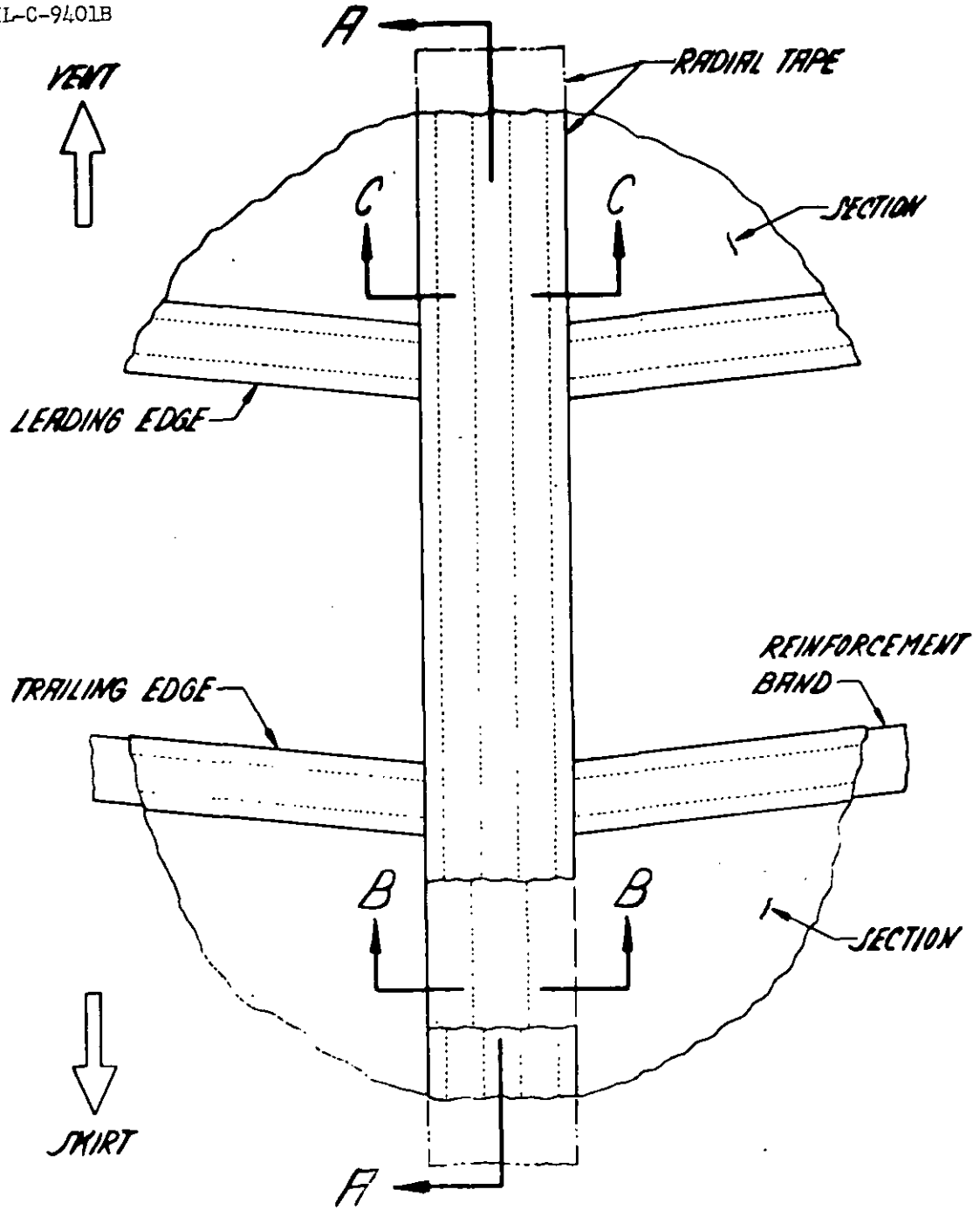


FIGURE 4. Connection of sections to radial tapes

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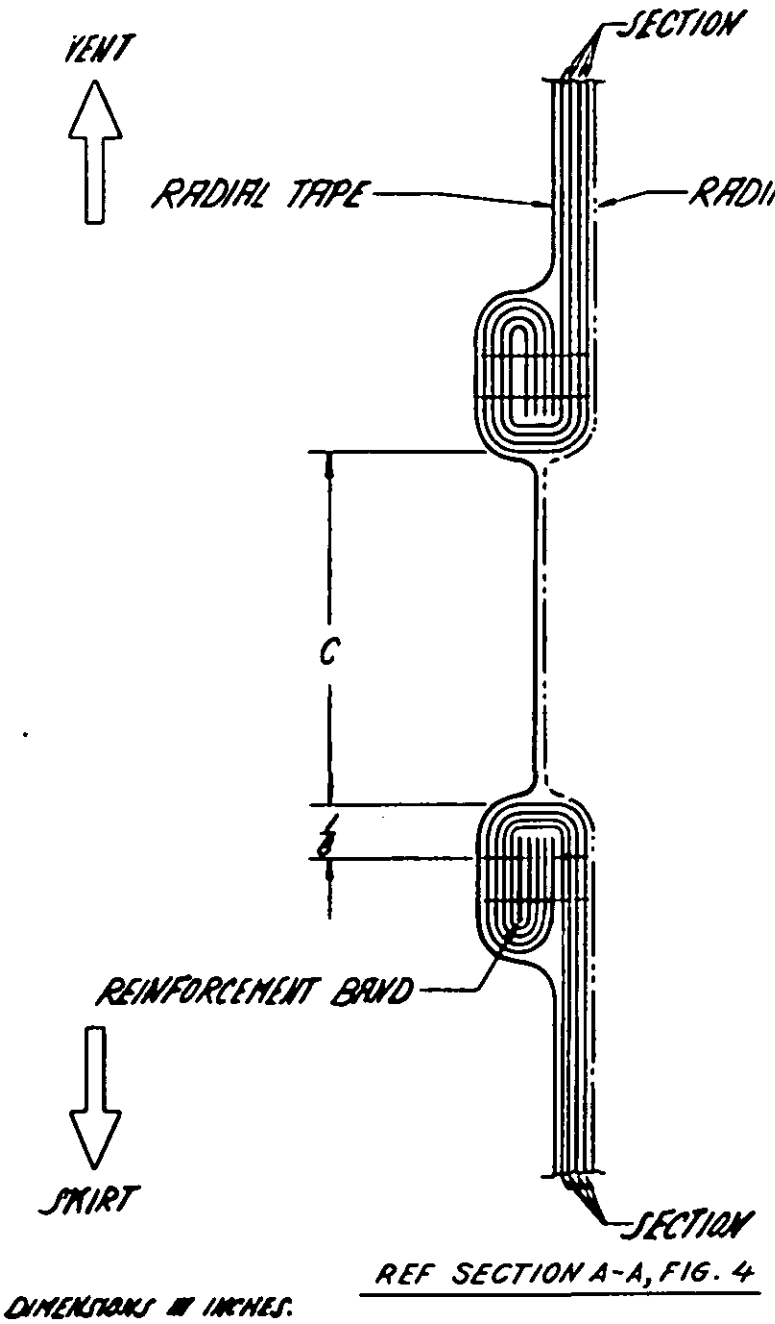


FIGURE 5. Hem details on section edges, style A

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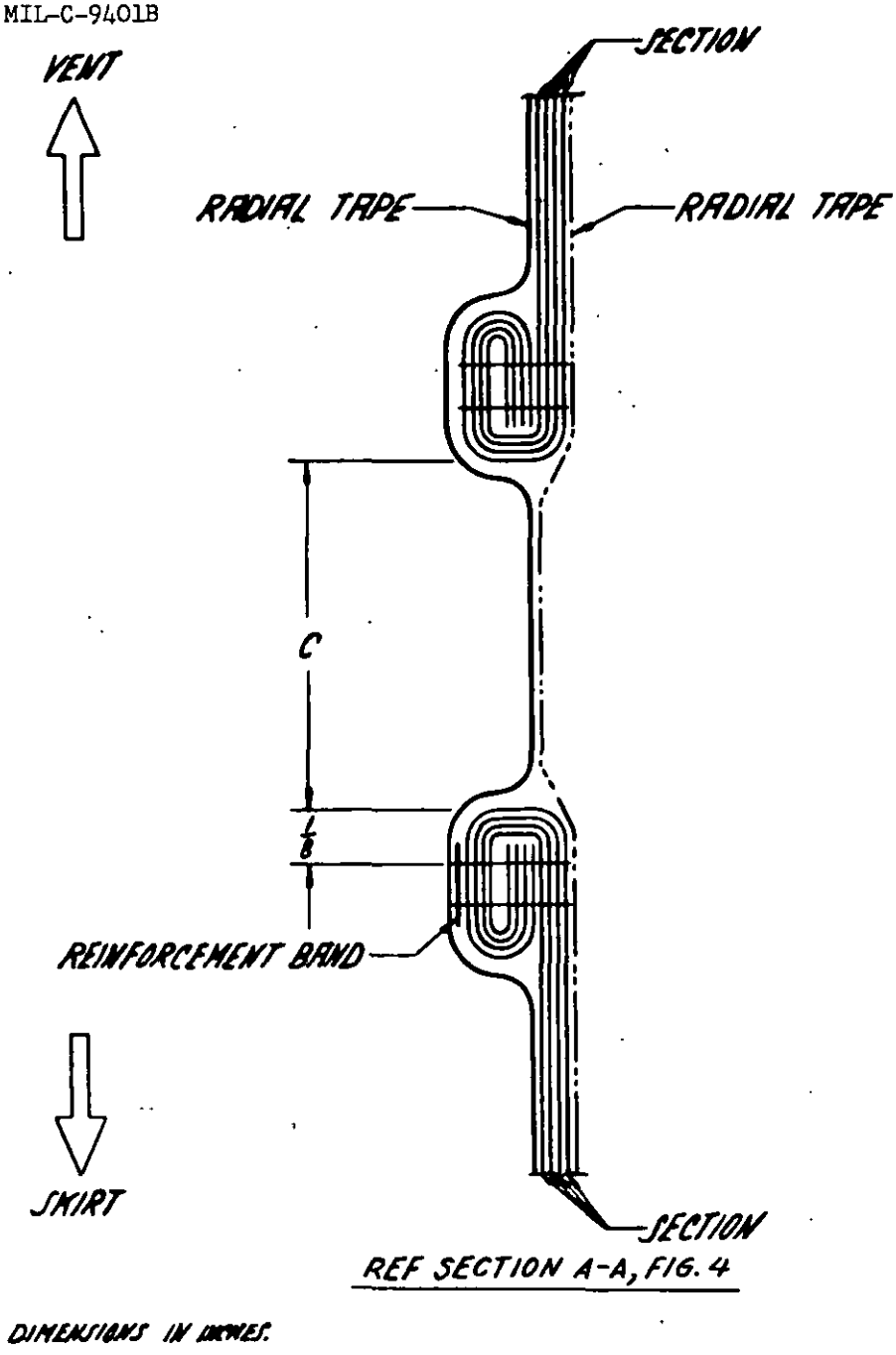
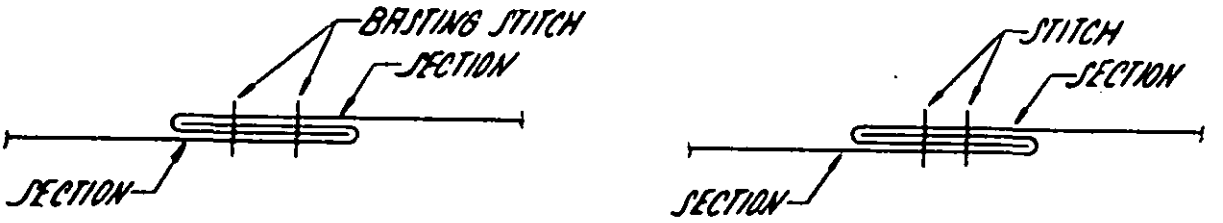


FIGURE 6. Hem details on section edges, style B

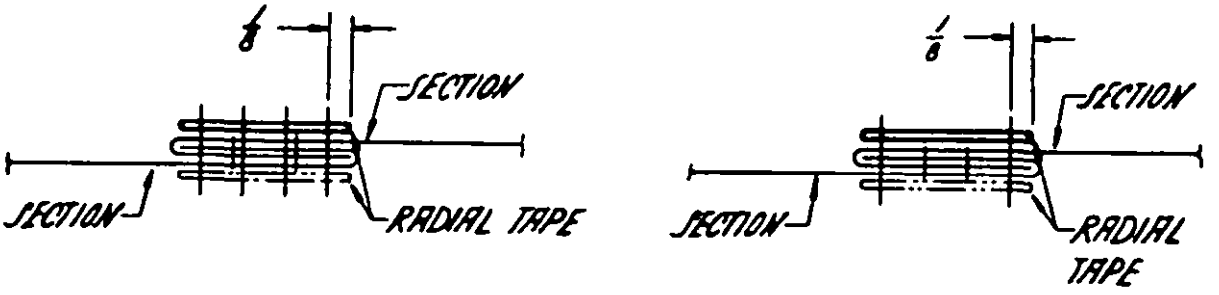
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STYLE A

STYLE B

METHOD OF JOINING SECTIONS
REF SECTION B-B, FIG. 4



STYLE A

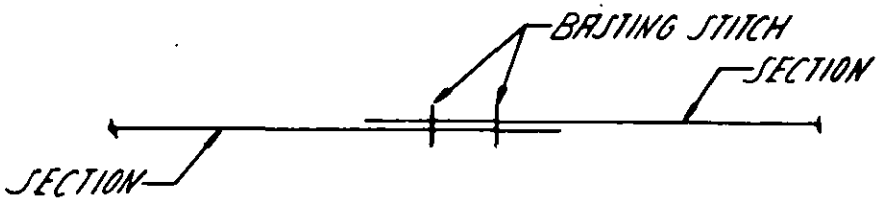
STYLE B

ATTACHMENT OF RADIAL TAPES
REF SECTION C-C, FIG. 4

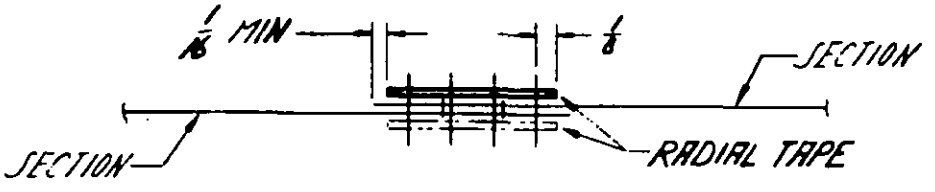
DIMENSIONS IN INCHES.

FIGURE 7. Radial seam construction

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METHOD OF JOINING SECTIONS
REF SECTION B-B, FIG. 4



ATTACHMENT OF RADIAL TAPES
REF SECTION C-C, FIG. 4

DIMENSIONS IN INCHES.

FIGURE 8. Radial seam construction, style C

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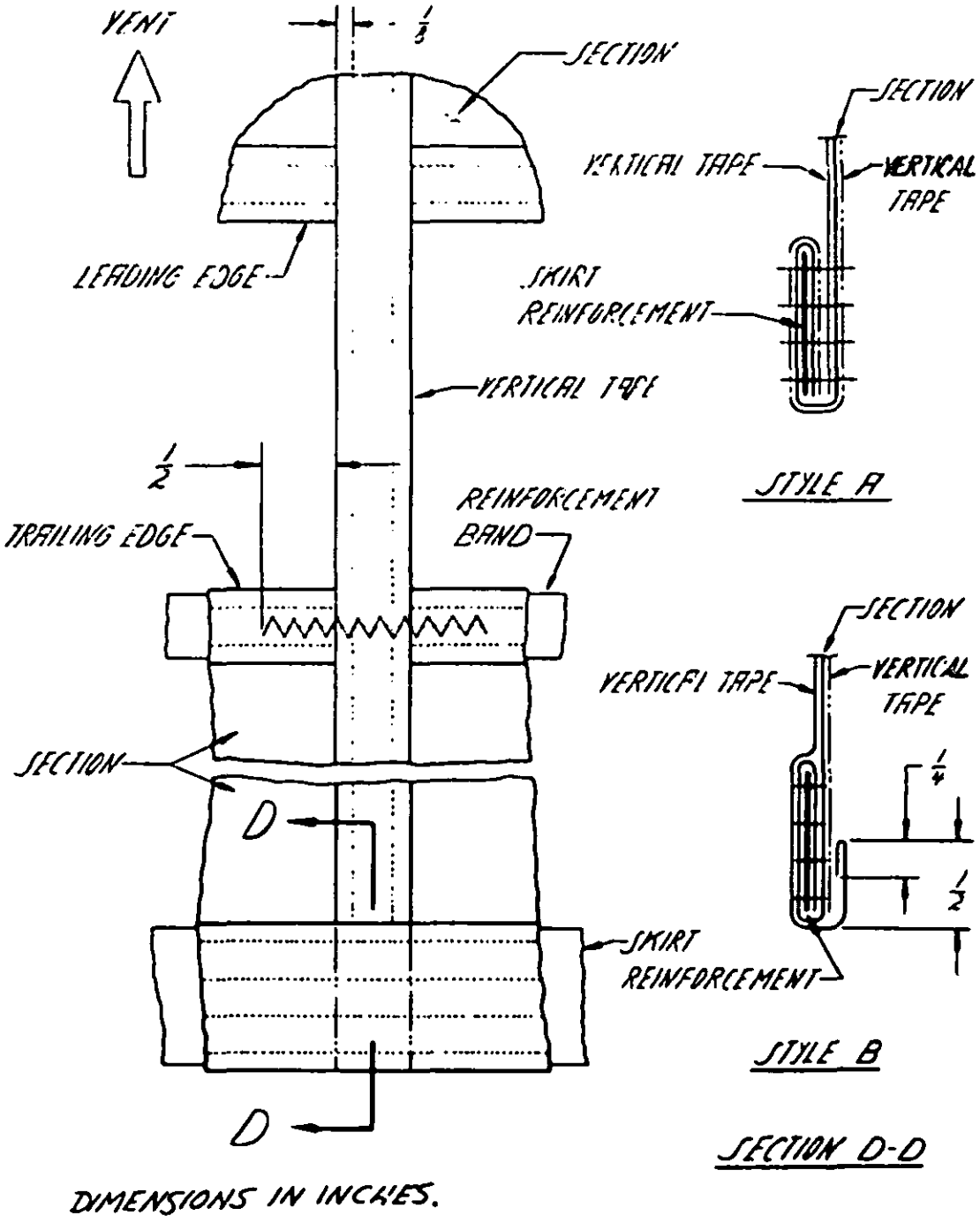


FIGURE 9. Vertical tape attachment at skirt area

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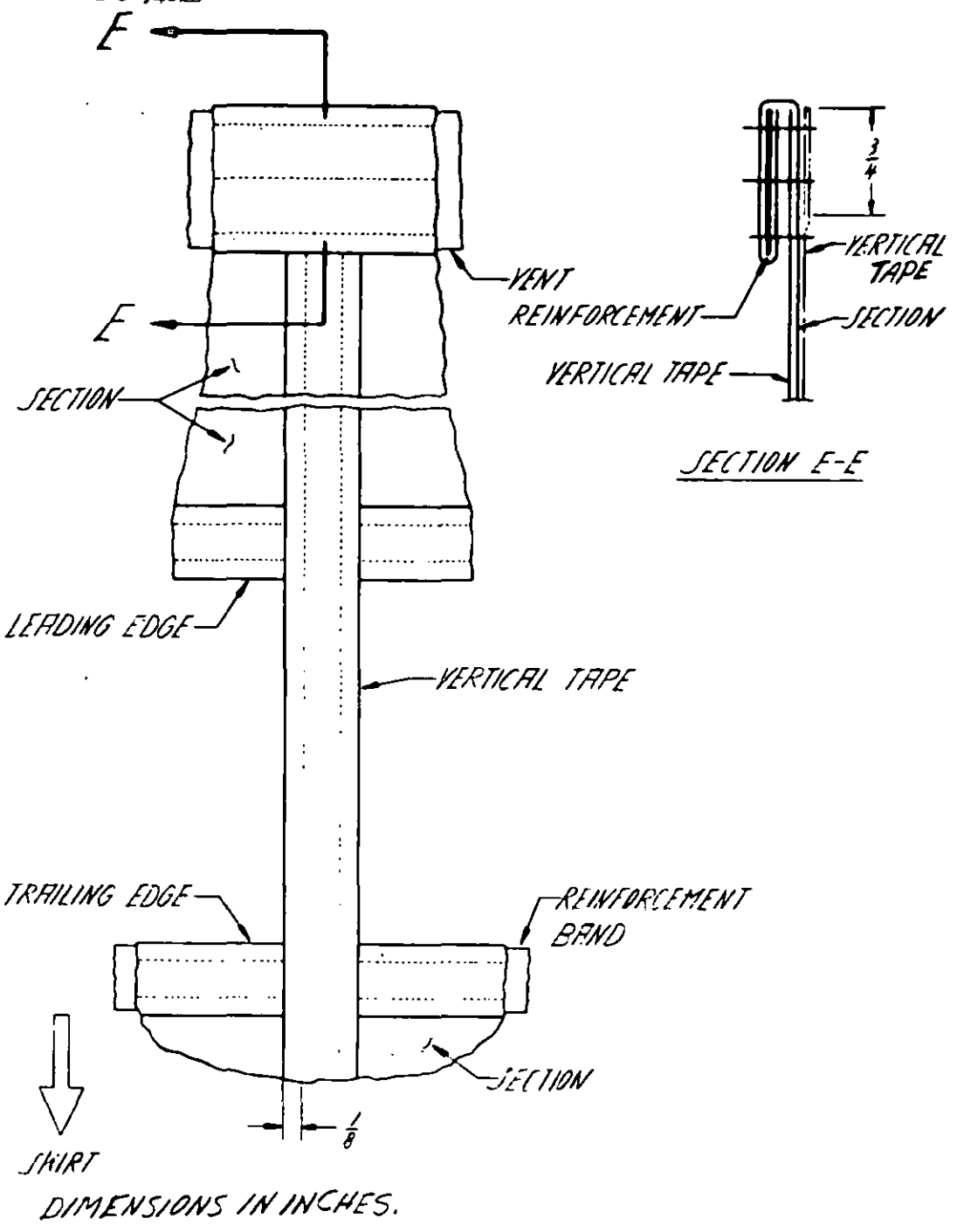
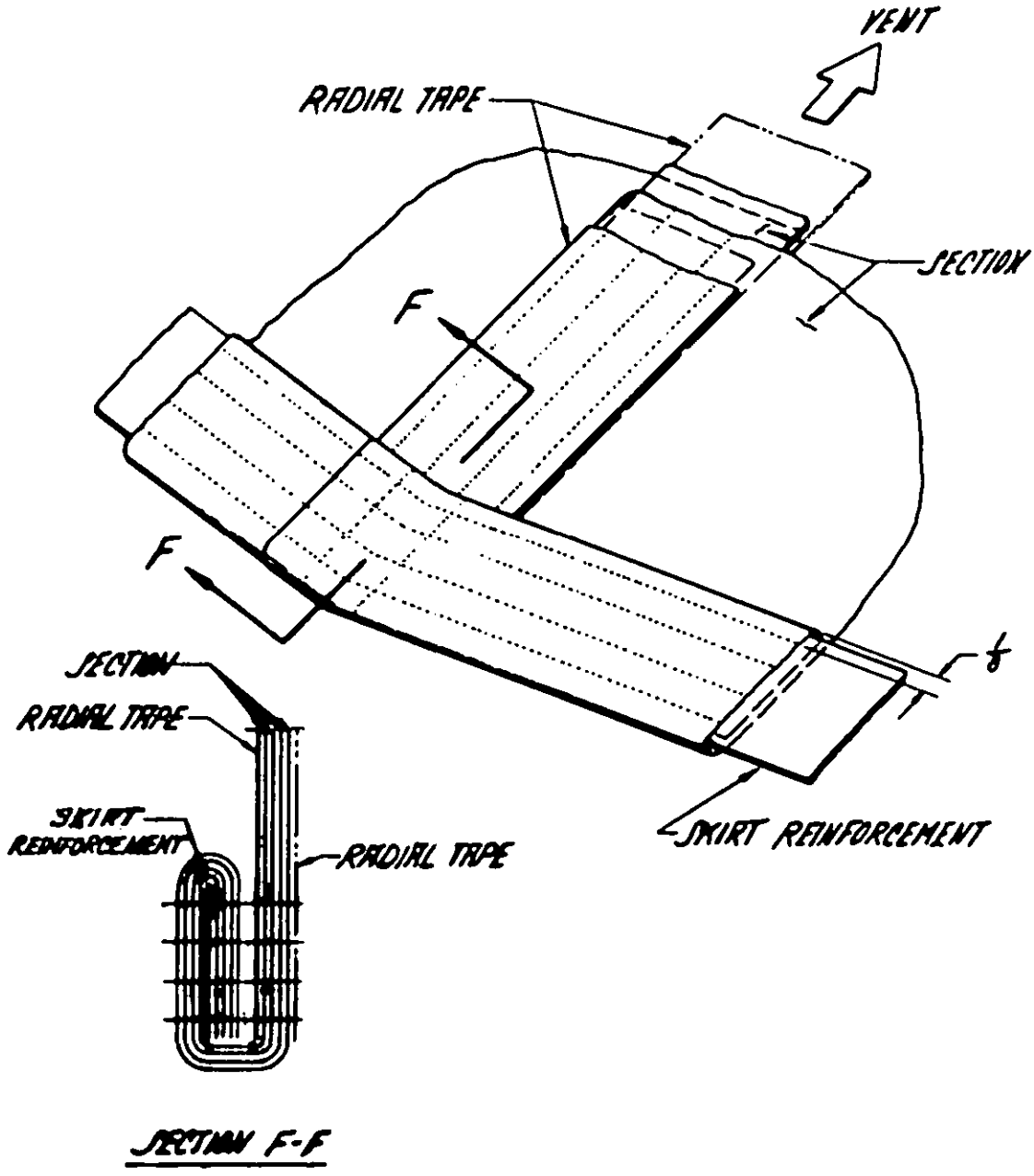


FIGURE 10. Vertical tape attachment at vent area

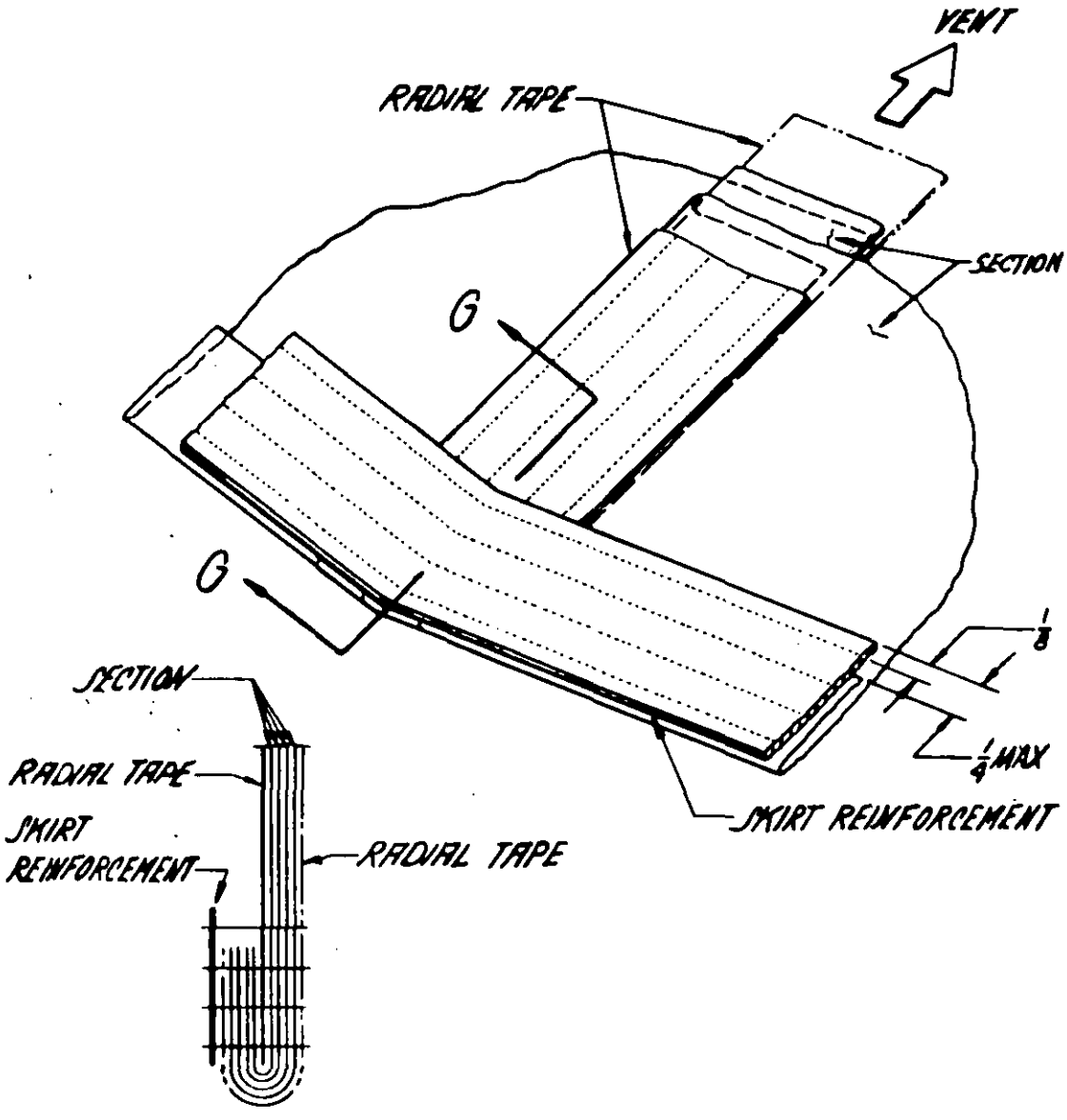
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DIMENSIONS IN INCHES.

FIGURE 11. Skirt construction at radial seam, style A

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DIMENSIONS IN INCHES.

FIGURE 12. Skirt construction at radial seam, style B

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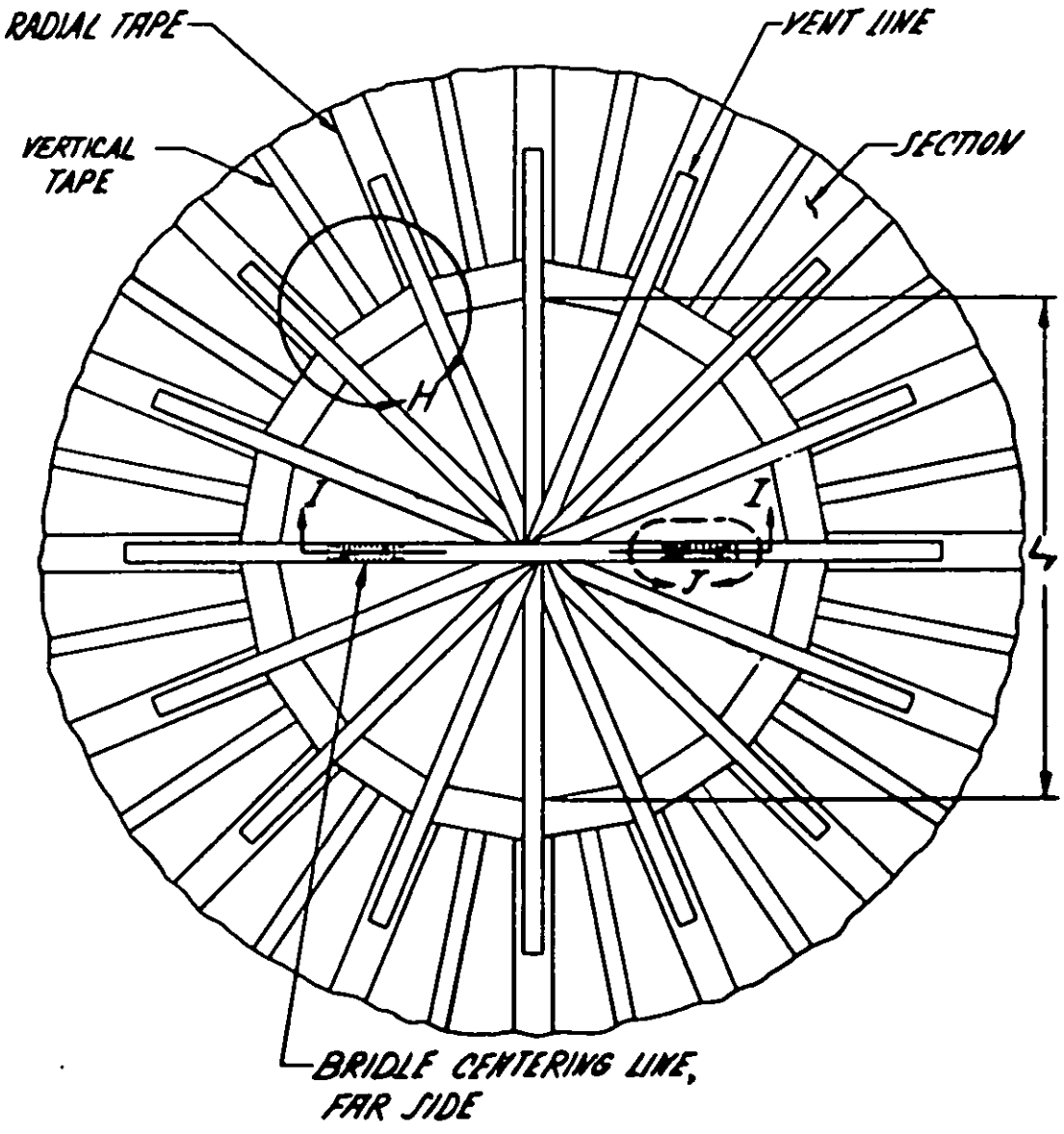
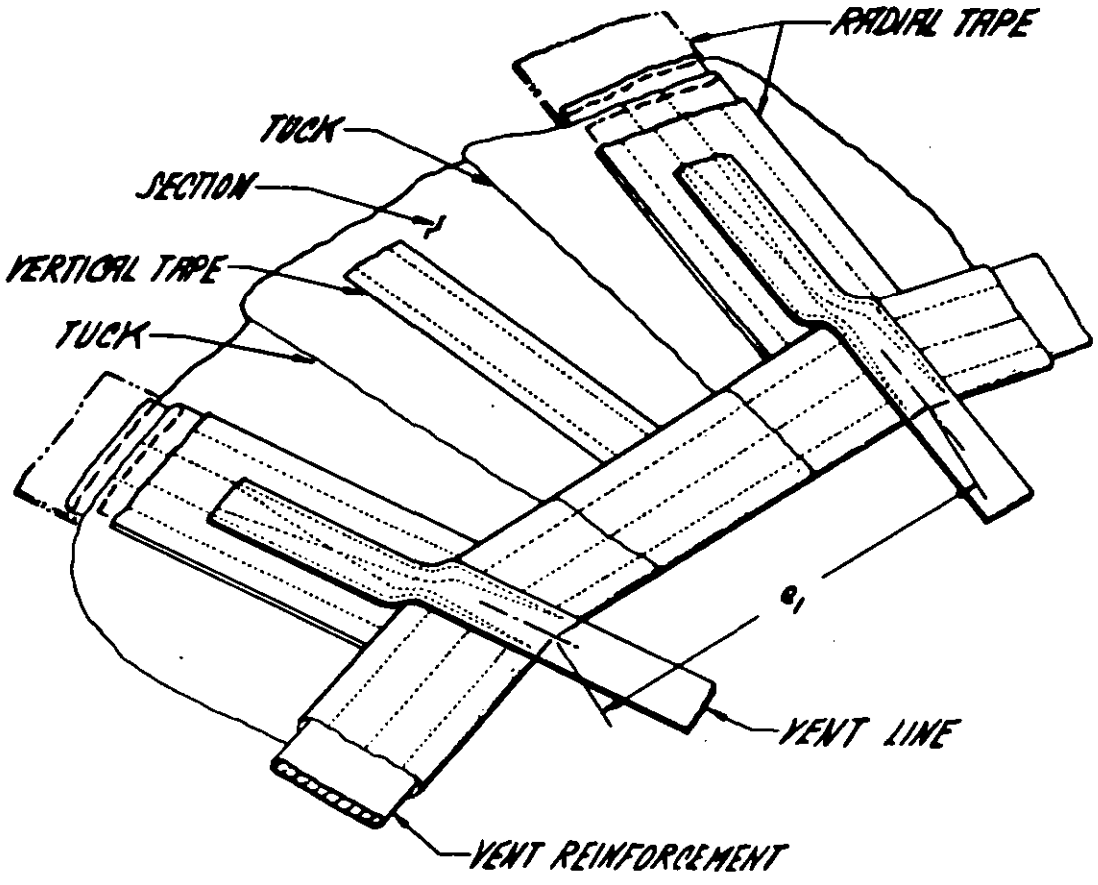


FIGURE 13. Top view of vent

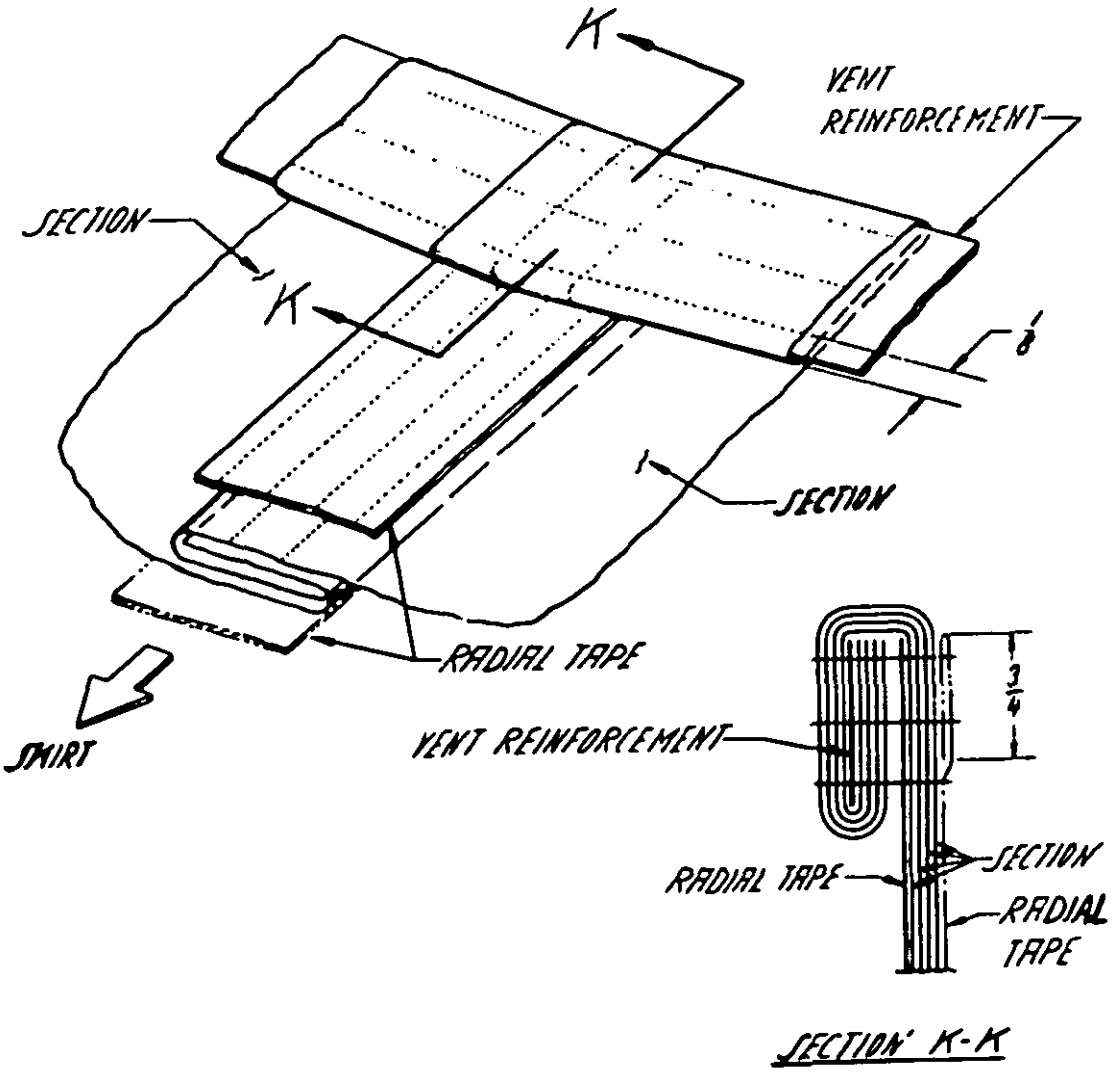
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REF VIEW H, FIG. 13

FIGURE 14. Details of vent

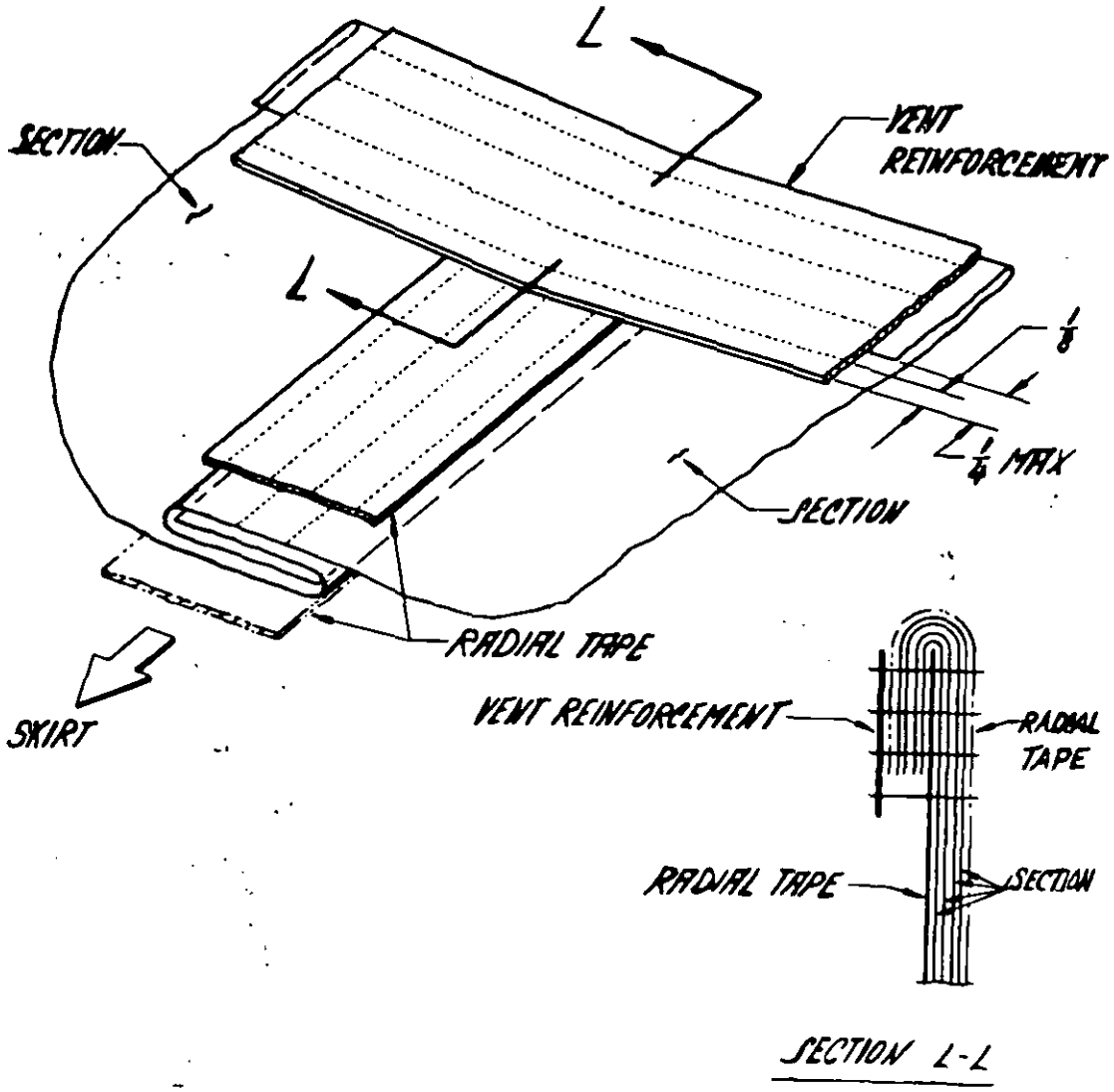
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DIMENSIONS IN INCHES.

FIGURE 15. Vent construction at radial seam, style A

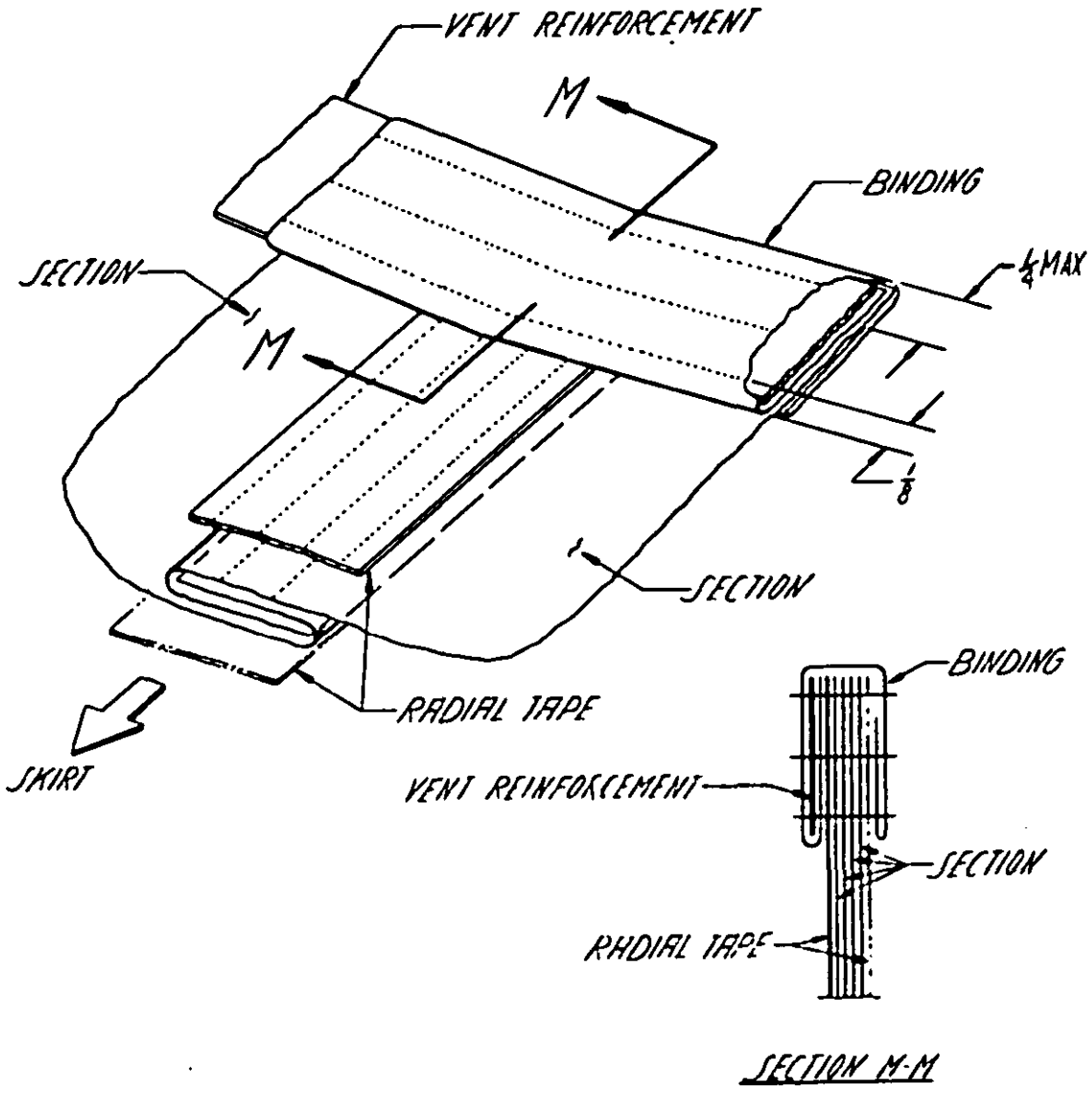
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DIMENSIONS IN INCHES.

FIGURE 16. Vent construction at radial seam, style B

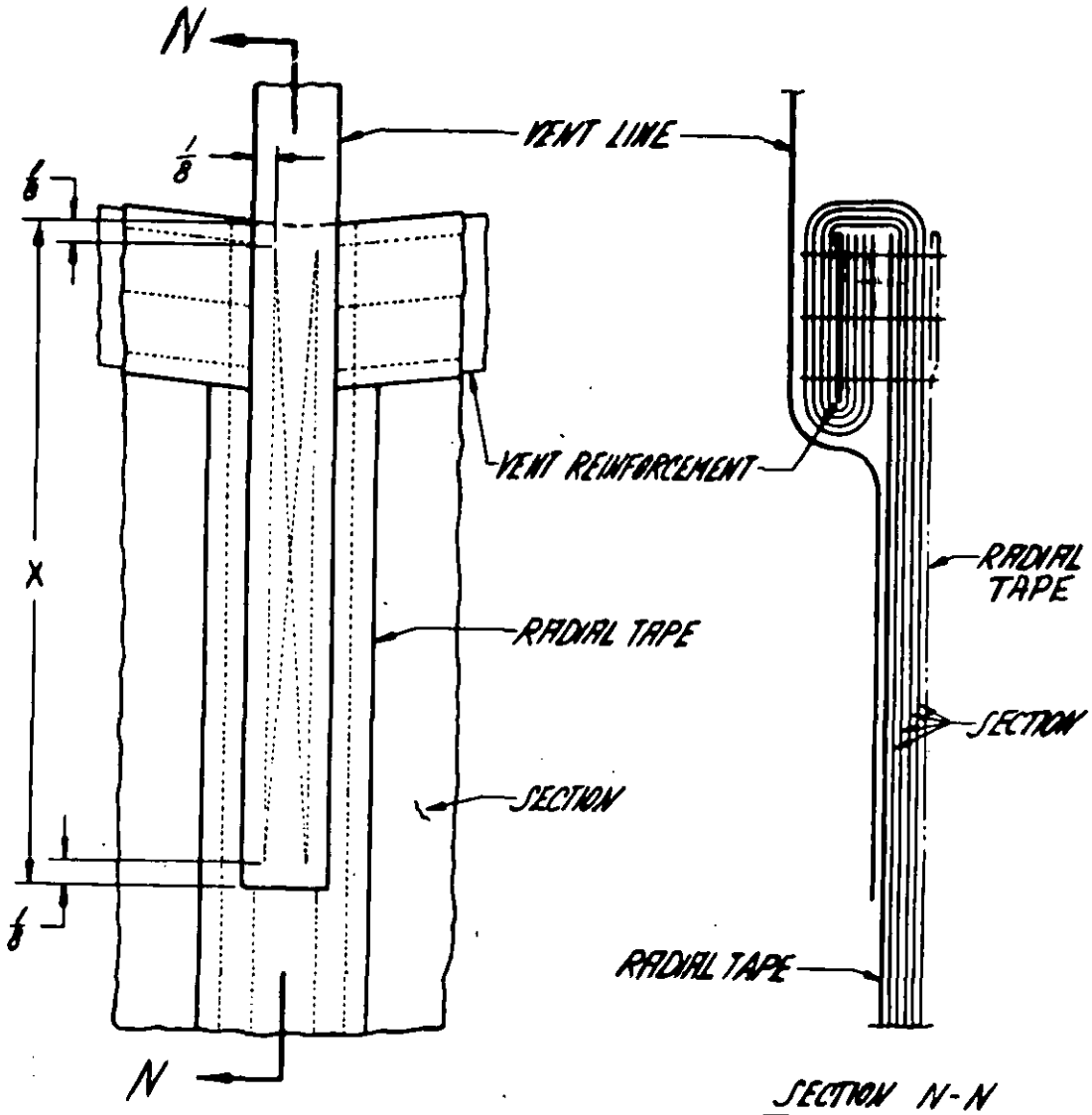
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DIMENSIONS IN INCHES.

FIGURE 17. Vent construction at radial seam, style C

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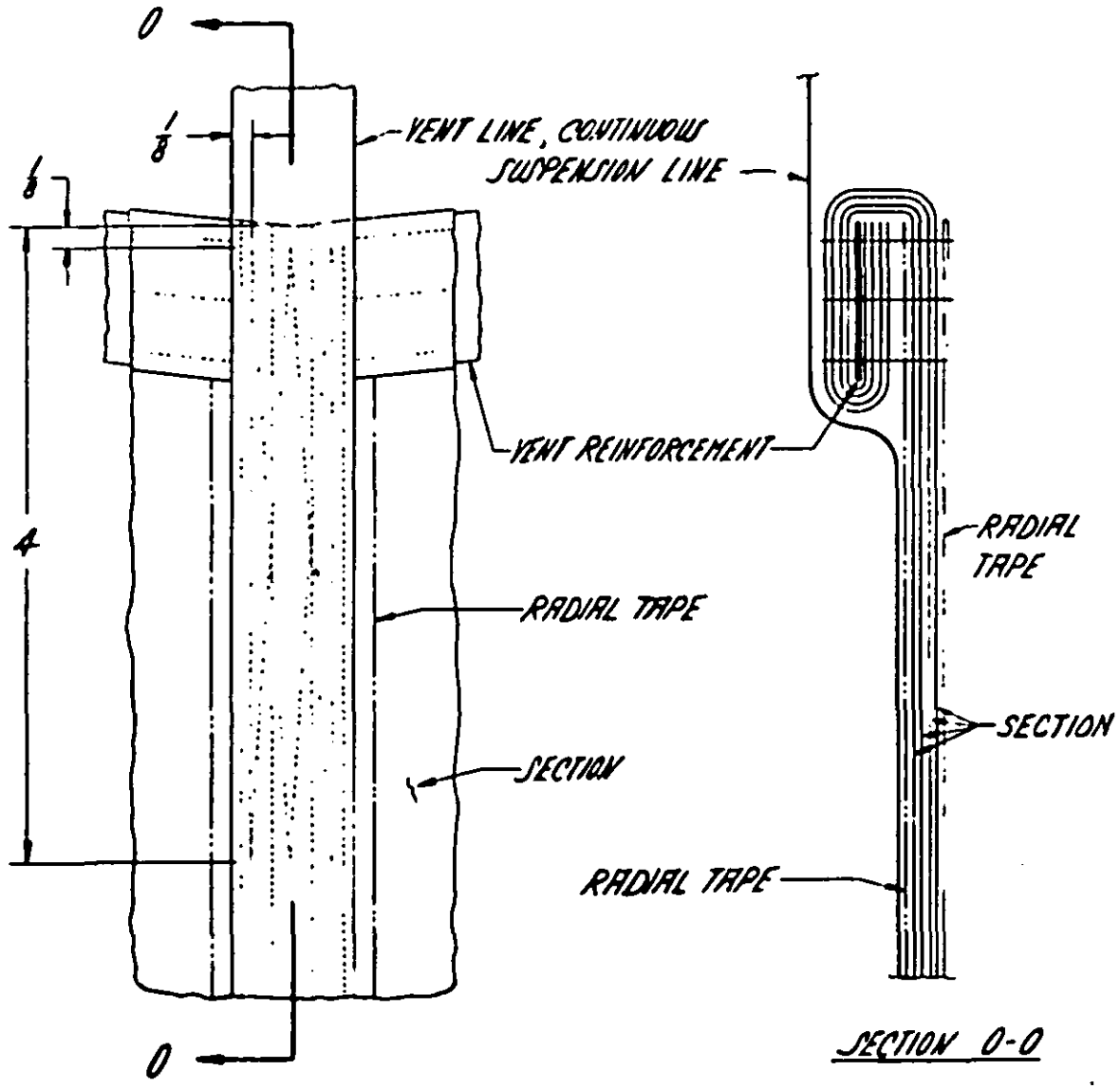


STRENGTH OF VENT LINE LB	X
LESS THAN 3,000	4
3,000 AND GREATER	6

DIMENSIONS IN INCHES.

FIGURE 18. Vent line attachment, style A

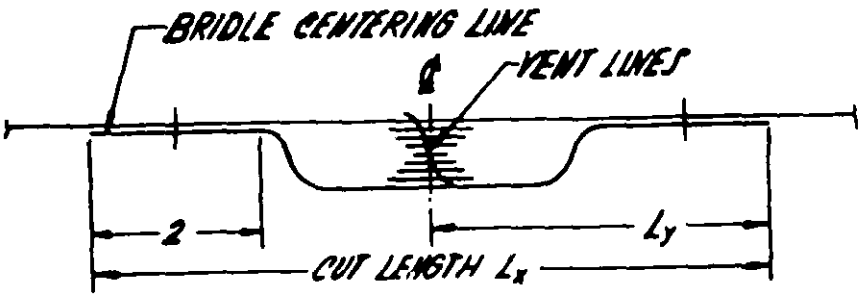
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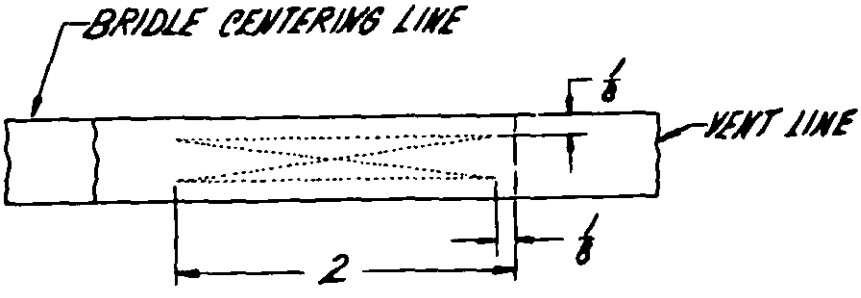
DIMENSIONS IN INCHES.

FIGURE 19. Vent line attachment, style B

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CONFIGURATION
REF SECTION I-I, FIG. 13

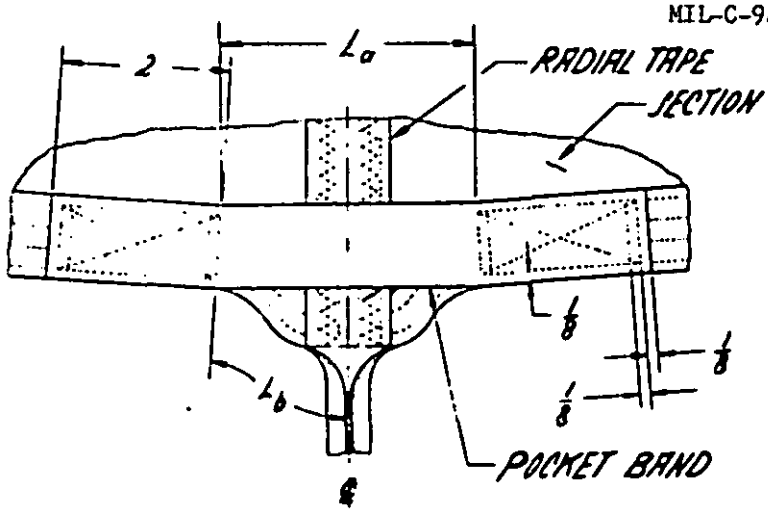


ATTACHMENT
REF VIEW J, FIG. 13

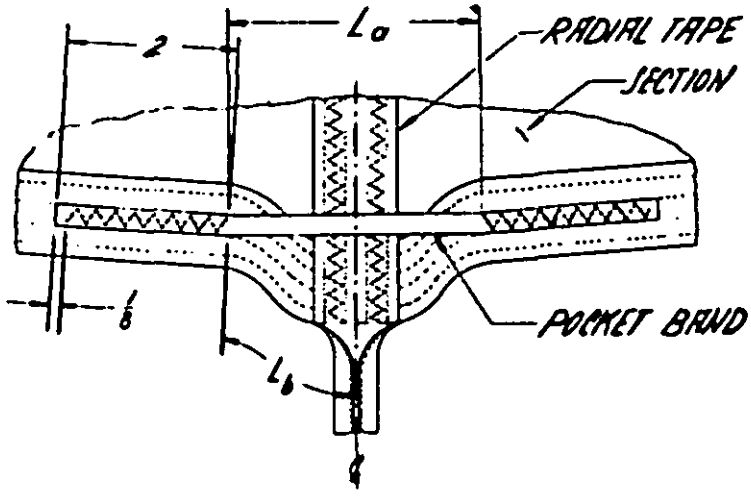
DIMENSIONS IN INCHES.

FIGURE 20. Details of bridle centering line

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STYLE A



STYLE B

DIMENSIONS IN INCHES.

FIGURE 21. Pocket band

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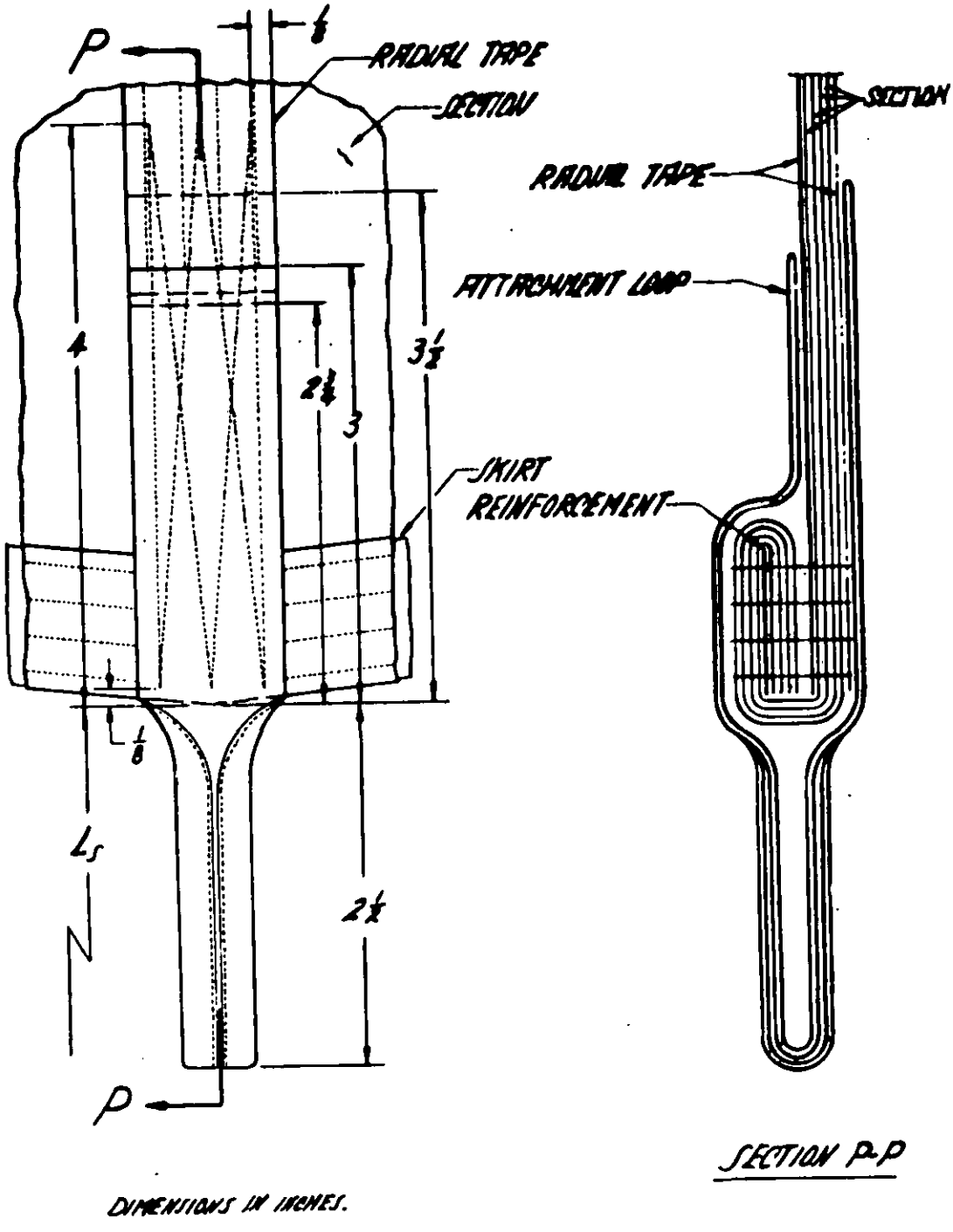


FIGURE 22. Suspension line attachment loop, style A

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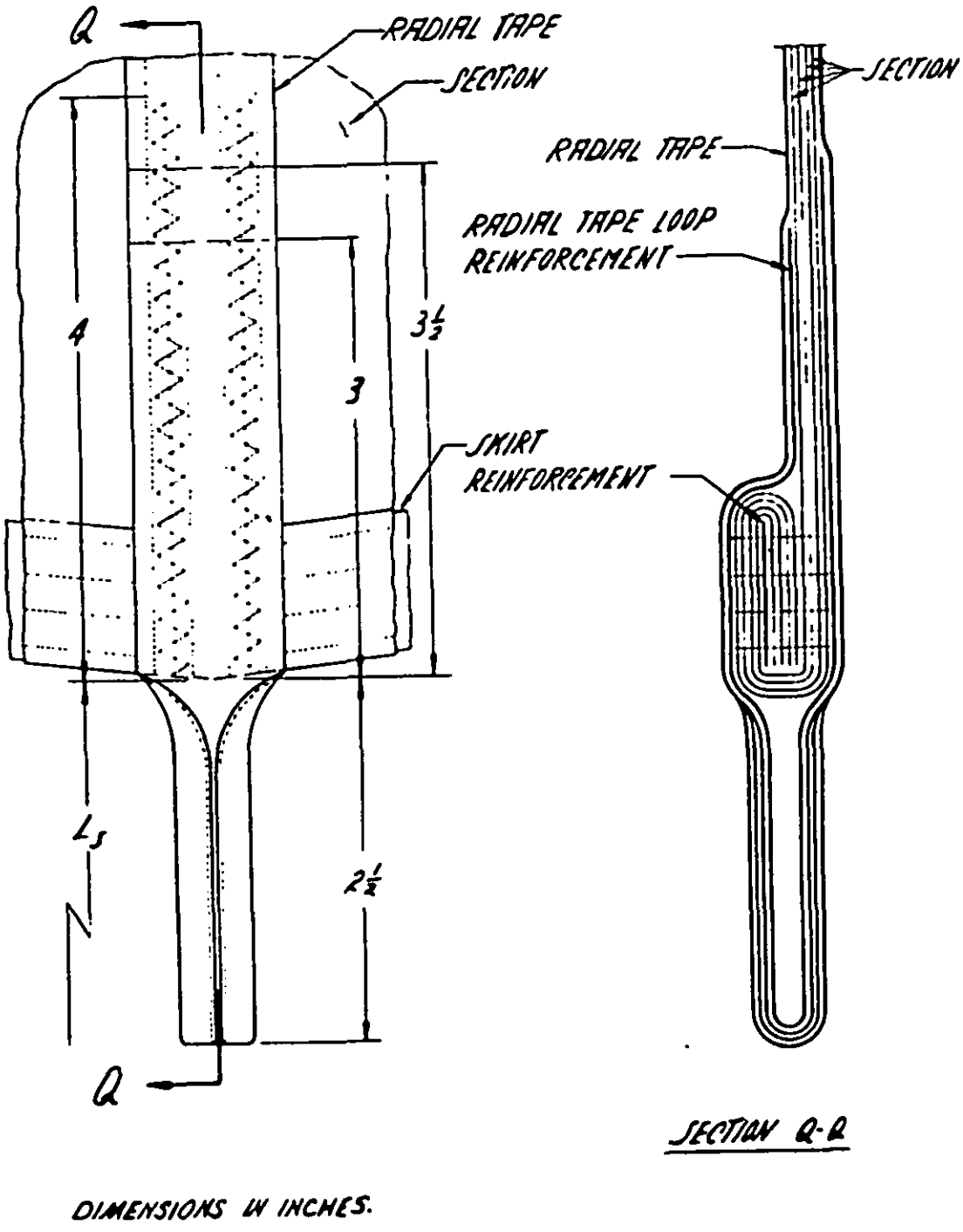


FIGURE 23. Suspension line attachment loop, style B

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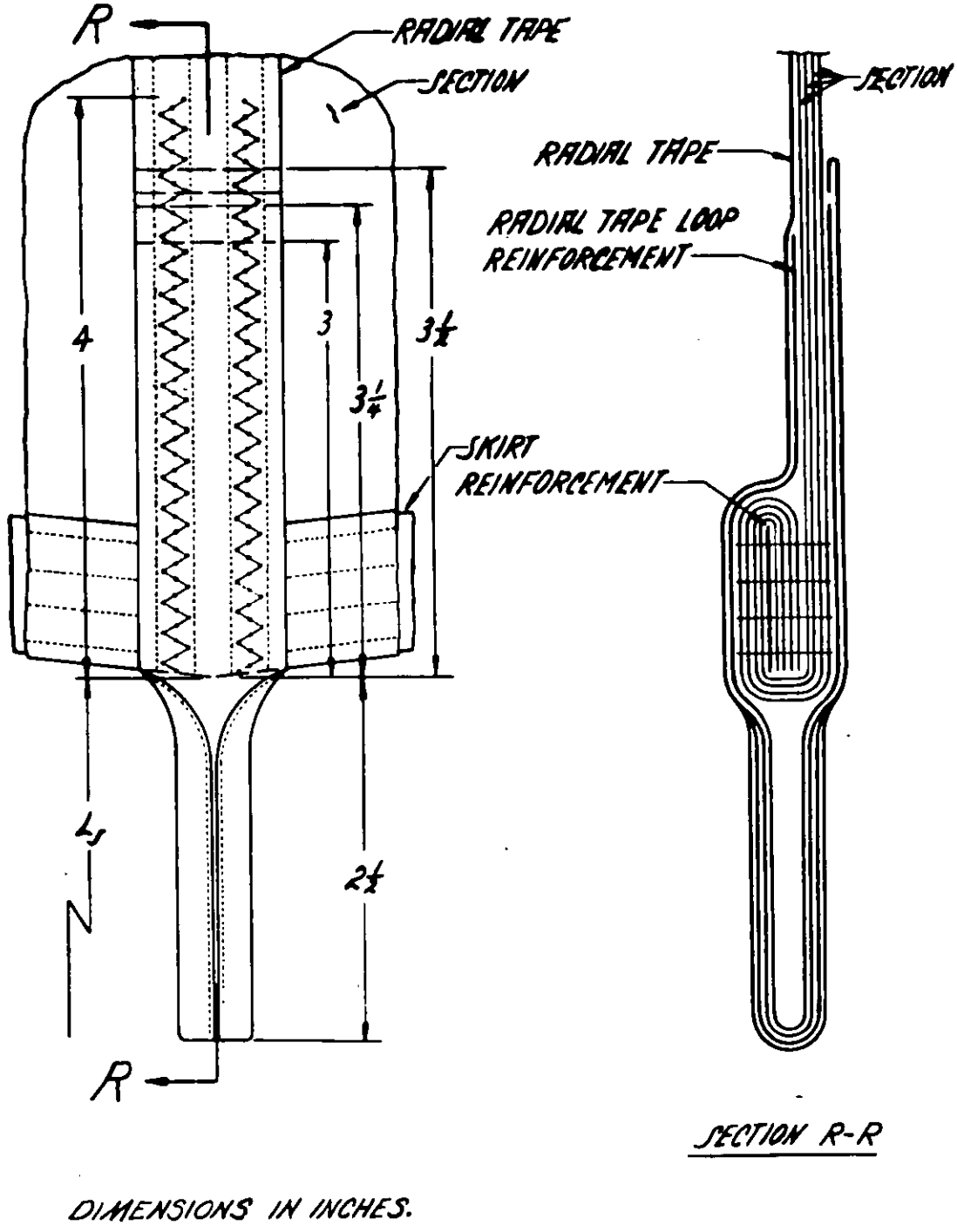
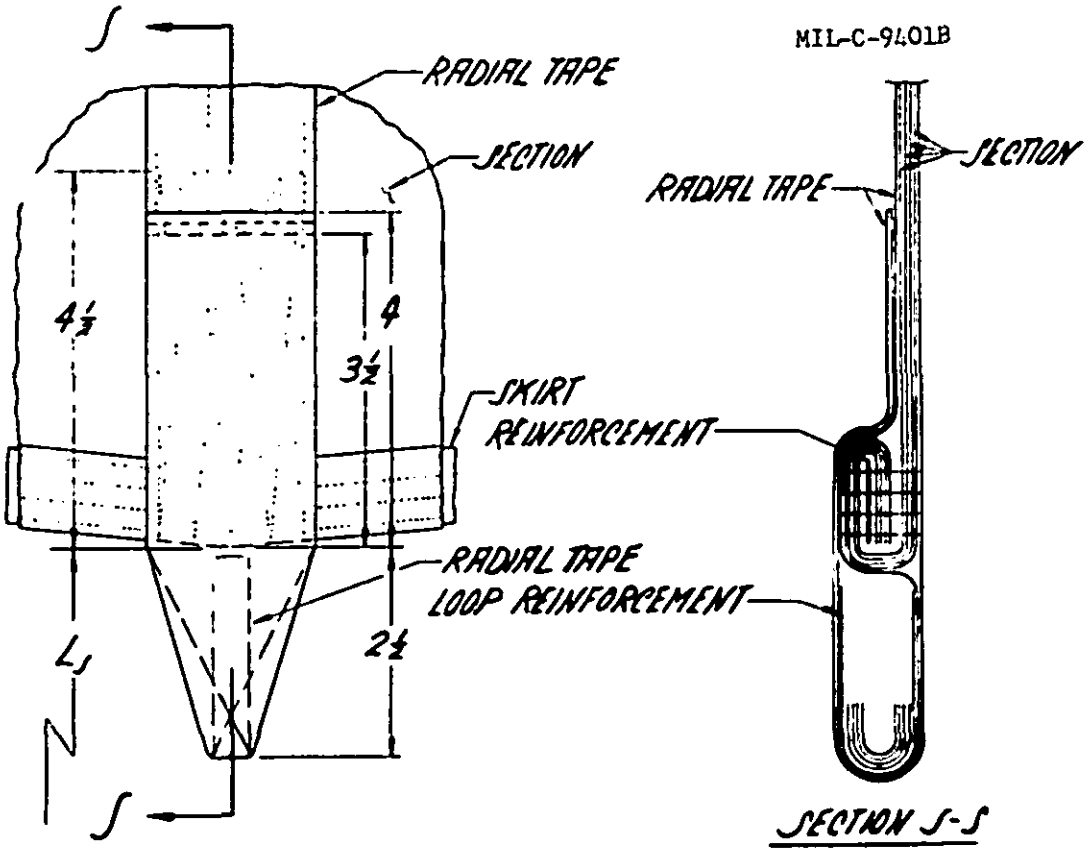
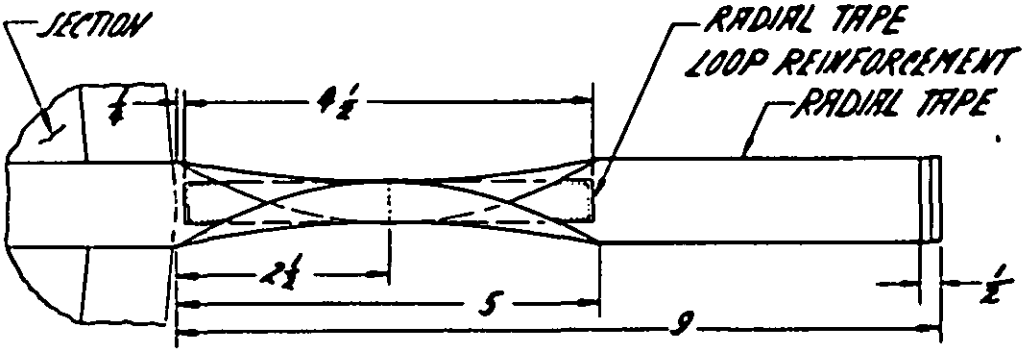


FIGURE 24. Suspension line attachment loop, style C

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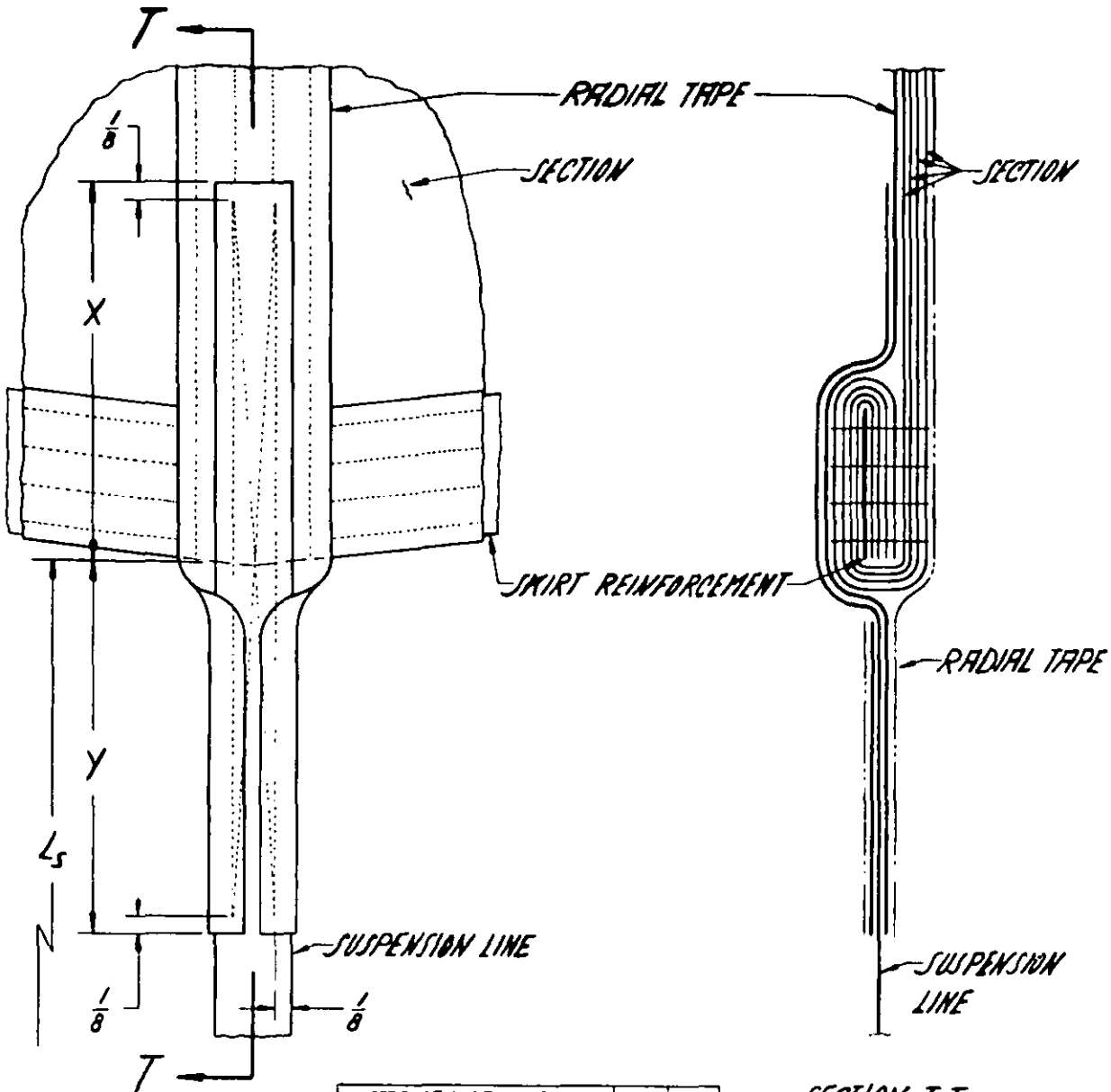


RADIAL TAPE ARRANGEMENT PRIOR TO FORMING LOOP

DIMENSIONS IN INCHES.

FIGURE 25. Suspension line attachment loop, style D

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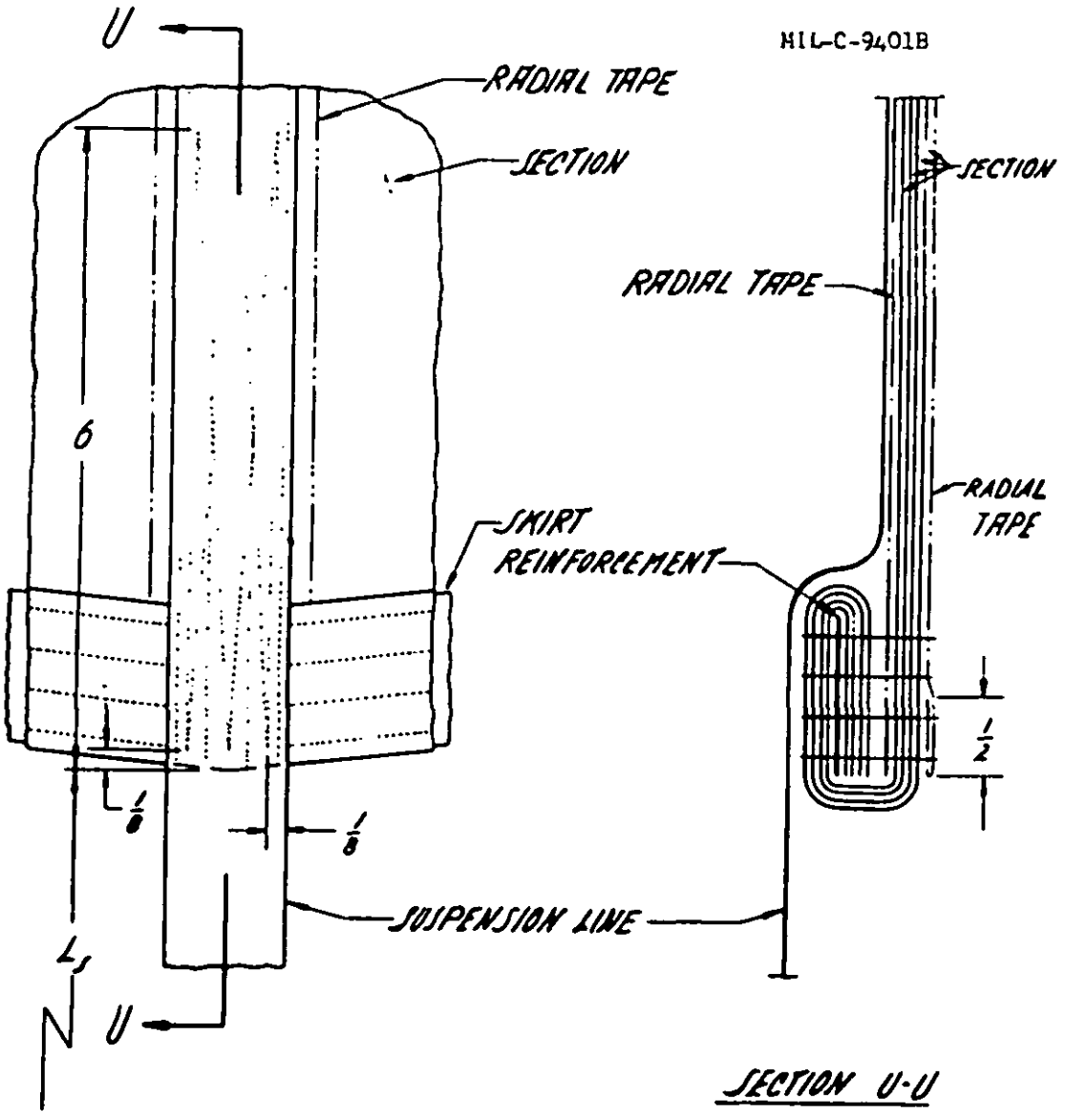


STRENGTH OF SUSPENSION LINE LB	X	Y
LESS THAN 3,000	4	4
3,000 AND GREATER	6	6

DIMENSIONS IN INCHES.

FIGURE 26. Suspension line attachment, style A

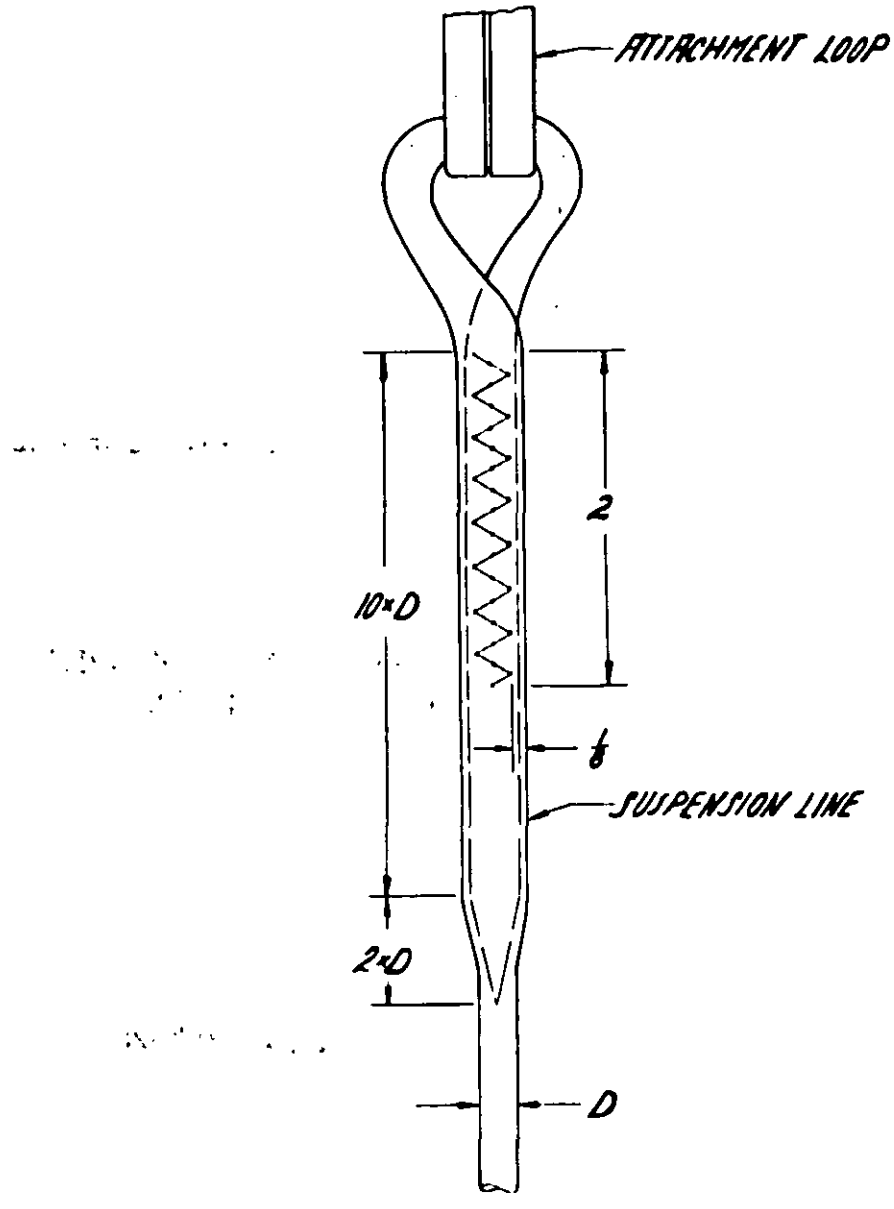
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DIMENSIONS IN INCHES.

FIGURE 27. Suspension line attachment, style B

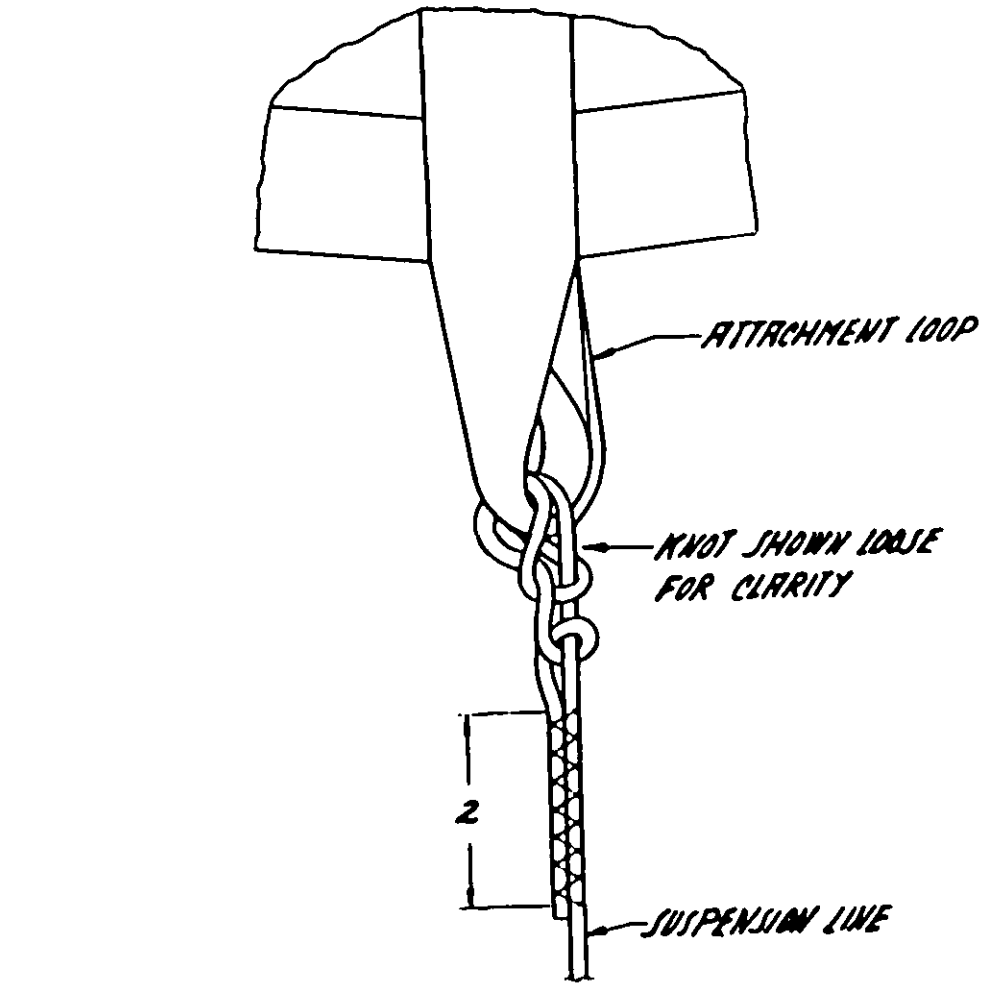
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DIMENSIONS IN INCHES.

FIGURE 28. Suspension line loop, style A

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DIMENSIONS IN INCHES.

FIGURE 29. Suspension line loop, style B

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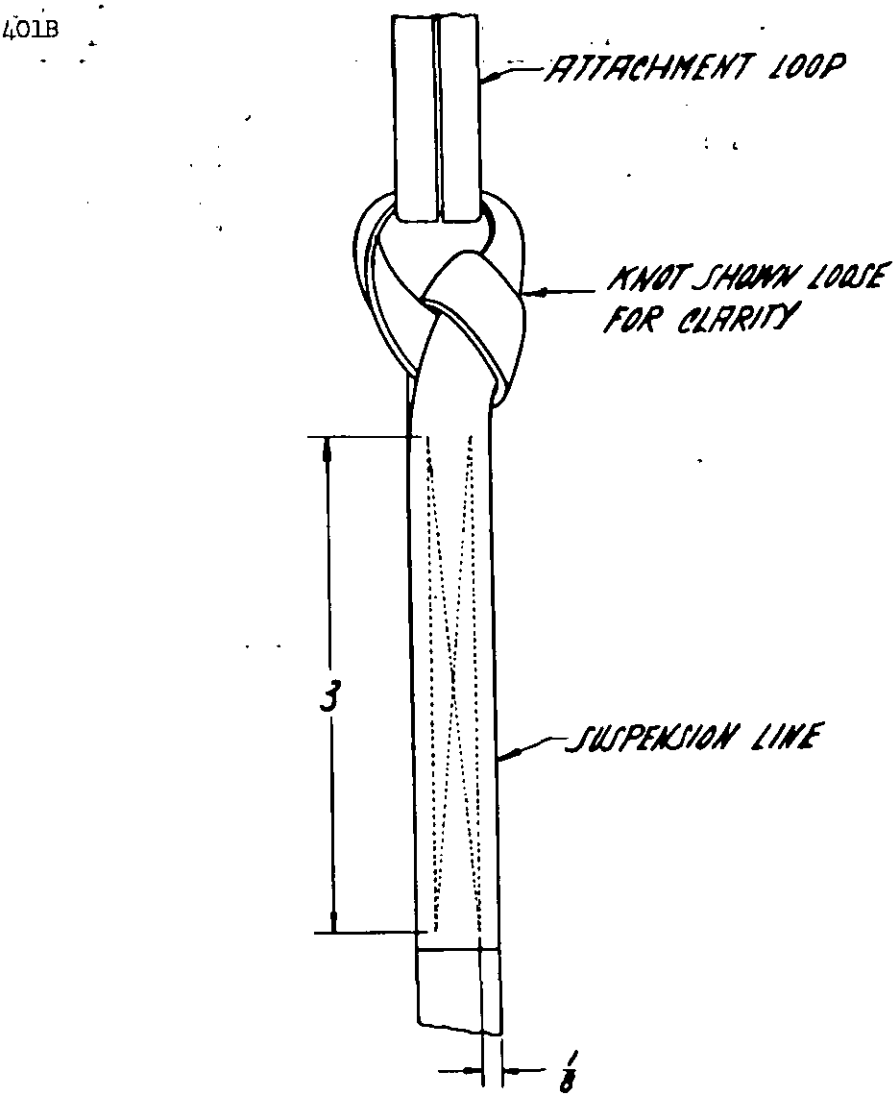


FIGURE 30. Suspension line loop, style C

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-C-9401B Canopy, Parachute, Ring Slot, Gen. Spec. for Constr. of

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

( ) Direct Government Contract ( ) Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?

( ) YES ( ) NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

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